DC servomotors. There are many types of dc motors in use in industries. DC motors that are used in servo systems are called dc servomotors. In dc servomotors, the rotor inertias have been made very small, with the result that motors with very high torque-to-inertia ratios are commercially available. Some dc servomotors have extremely small time constants. DC servomotors with relatively small power ratings are used in instruments and computer-related equipment such as disk drives, tape drives, printers, and word processors. DC servomotors with medium and large power ratings are used in robot systems, numerically controlled milling machines, and so on.

In dc servomotors, the field windings may be connected in series with the armature or the field windings may be separate from the armature. (That is, the magnetic field is produced by a separate circuit.) In the latter case, where the field is excited separately, the magnetic flux is independent of the armature current. In some dc servomotors, the magnetic field is produced by a permanent magnet and, therefore, the magnetic flux is constant. Such dc servomotors are called permanent magnet dc servomotors. DC servomotors with separately excited fields, as well as permanent magnet dc servomotors, can be controlled by the armature current. Such a scheme to control the output of the dc servomotor by the armature current is called armature control of dc servomotors.

In the case where the armature current is maintained constant and the speed is controlled by the field voltage, the dc motor is called a field-controlled dc motor. (Some speed control systems use field-controlled dc motors.) The requirement of constant armature current, however, is a serious disadvantage. (Providing a constant current source is much more difficult than providing a constant voltage source.) The time constants of the field-controlled dc motor are generally large compared with the time constants of a comparable armature-controlled dc motor.

A dc servomotor may also be driven by an electronic motion controller, frequently called a servodriver, as a motor–driver combination. The servodriver controls the motion of a dc servomotor and operates in various modes. Some of the features are point-to-point positioning, velocity profiling, and programmable acceleration. The use of an electronic motion controller using a pulse-width-modulated driver to control a dc servomotor is frequently seen in robot control systems, numerical control systems, and other position and/or speed control systems.

In what follows we shall discuss armature control of dc servomotors.

A servo system. Consider the servo system shown in Figure 4–8(a). The objective of this system is to control the position of the mechanical load in accordance with the reference position. The operation of this system is as follows: A pair of potentiometers acts as an error-measuring device. They convert the input and output positions into proportional electric signals. The command input signal determines the angular position r of the wiper arm of the input potentiometer. The angular position r is the reference input to the system, and the electric potential of the arm is proportional to the angular position of the arm. The output shaft position determines the angular position c of the wiper arm of the output potentiometer. The difference between the input angular position r and the output angular position c is the error signal e, or

$$e = r - c$$

The potential difference $e_r - e_c = e_v$ is the error voltage, where e_r is proportional to r and e_c is proportional to c; that is, $e_r = K_0 r$ and $e_c = K_0 c$, where K_0 is a proportionality

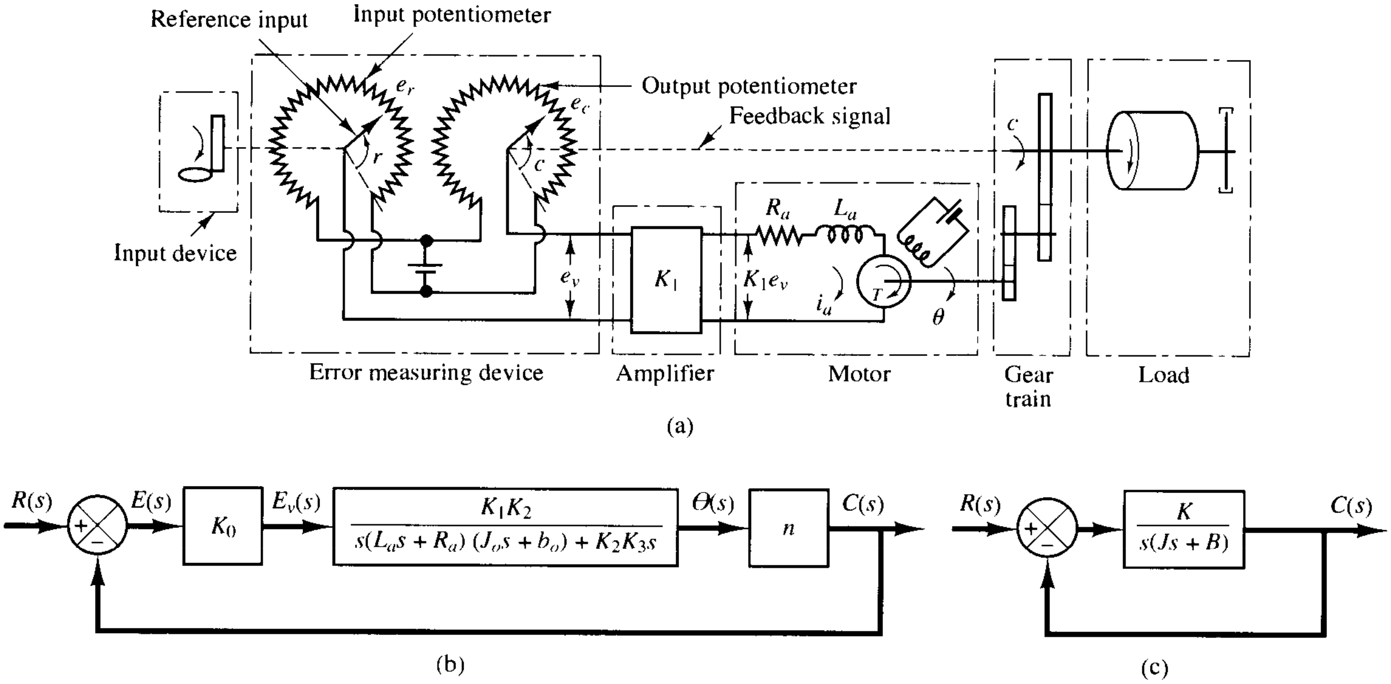


Figure 4-8
(a) Schematic diagram of servo system; (b) block diagram for the system; (c) simplified block diagram.

constant. The error voltage that appears at the potentiometer terminals is amplified by the amplifier whose gain constant is K_1. The output voltage of this amplifier is applied to the armature circuit of the dc motor. (The amplifier must have very high input impedance because the potentiometers are essentially high impedance circuits and do not tolerate current drain. At the same time, the amplifier must have low output impedance since it feeds into the armature circuit of the motor.) A fixed voltage is applied to the field winding. If an error exists, the motor develops a torque to rotate the output load in such a way as to reduce the error to zero. For constant field current, the torque developed by the motor is

$$T = K_2 i_a$$

where K_2 is the motor torque constant and i_a is the armature current.

Notice that if the sign of the current i_a is reversed the sign of the torque T will be reversed, which will result in the reversion of the direction of rotor rotation.

When the armature is rotating, a voltage proportional to the product of the flux and angular velocity is induced in the armature. For a constant flux, the induced voltage e_b is directly proportional to the angular velocity $d\theta/dt$, or

$$e_b = K_3 \frac{d\theta}{dt} \tag{4–9}$$

where e_b is the back emf, K_3 is the back emf constant of the motor, and θ is the angular displacement of the motor shaft.

The speed of an armature-controlled dc servomotor is controlled by the armature voltage e_a. (The armature voltage $e_a = K_1 e_v$ is the output of the amplifier.) The differential equation for the armature circuit is

$$L_a \frac{di_a}{dt} + R_a i_a + e_b = e_a$$

or

$$L_a \frac{di_a}{dt} + R_a i_a + K_3 \frac{d\theta}{dt} = K_1 e_v \tag{4–10}$$

The equation for torque equilibrium is

$$J_0 \frac{d^2\theta}{dt^2} + b_0 \frac{d\theta}{dt} = T = K_2 i_a \tag{4–11}$$

where J_0 is the inertia of the combination of the motor, load, and gear train referred to the motor shaft and b_0 is the viscous-friction coefficient of the combination of the motor, load, and gear train referred to the motor shaft. The transfer function between the motor shaft displacement and the error voltage is obtained from Equations (4–10) and (4–11) as follows:

$$\frac{\Theta(s)}{E_v(s)} = \frac{K_1 K_2}{s(L_a s + R_a)(J_0 s + b_0) + K_2 K_3 s} \tag{4–12}$$

where $\Theta(s) = \mathcal{L}[\theta(t)]$ and $E_v(s) = \mathcal{L}[e_v(t)]$. We assume that the gear ratio of the gear train is such that the output shaft rotates n times for each revolution of the motor shaft. Thus,

$$C(s) = n\Theta(s) \tag{4–13}$$

where $C(s) = \mathcal{L}[c(t)]$ and $c(t)$ is the angular displacement of the output shaft. The relationship among $E_v(s)$, $R(s)$, and $C(s)$ is

$$E_v(s) = K_0[R(s) - C(s)] = K_0 E(s) \tag{4–14}$$

where $R(s) = \mathcal{L}[r(t)]$. The block diagram of this system can be constructed from Equations (4–12), (4–13), and (4–14), as shown in Figure 4–8(b). The transfer function in the feedforward path of this system is

$$G(s) = \frac{C(s)}{\Theta(s)} \frac{\Theta(s)}{E_v(s)} \frac{E_v(s)}{E(s)} = \frac{K_0 K_1 K_2 n}{s[(L_a s + R_a)(J_0 s + b_0) + K_2 K_3]}$$

Since L_a is usually small, it can be neglected, and the transfer function $G(s)$ in the feedforward path becomes

$$\begin{aligned} G(s) &= \frac{K_0 K_1 K_2 n}{s[R_a(J_0 s + b_0) + K_2 K_3]} \\ &= \frac{K_0 K_1 K_2 n / R_a}{J_0 s^2 + \left(b_0 + \frac{K_2 K_3}{R_a}\right) s} \end{aligned} \tag{4–15}$$

The term $[b_0 + (K_2K_3/R_a)]s$ indicates that the back emf of the motor effectively increases the viscous friction of the system. The inertia J_0 and viscous friction coefficient $b_0 + (K_2K_3/R_a)$ are referred to the motor shaft. When J_0 and $b_0 + (K_2K_3/R_a)$ are multiplied by $1/n^2$, the inertia and viscous-friction coefficient are expressed in terms of the output shaft. Introducing new parameters defined by

$J = J_0/n^2$ = moment of inertia referred to the output shaft

$B = [b_0 + (K_2K_3/R_a)]/n^2$ = viscous-friction coefficient referred to the output shaft

$K = K_0K_1K_2/nR_a$

the transfer function $G(s)$ given by Equation (4–15) can be simplified, yielding

$$G(s) = \frac{K}{Js^2 + Bs}$$

or

$$G(s) = \frac{K_m}{s(T_m s + 1)} \tag{4–16}$$

where

$$K_m = \frac{K}{B}, \qquad T_m = \frac{J}{B} = \frac{R_a J_0}{R_a b_0 + K_2 K_3}$$

The block diagram of the system shown in Figure 4–8(b) can thus be simplified as shown in Figure 4–8(c).

In the following, we shall investigate the dynamic responses of this system to unit-step, unit-ramp, and unit-impulse inputs.

From Equations (4–15) and (4–16), it can be seen that the transfer functions involve the term $1/s$. Thus, this system possesses an integrating property. In Equation (4–16), notice that the time constant of the motor is smaller for a smaller R_a and smaller J_0. With small J_0, as the resistance R_a is reduced, the motor time constant approaches zero, and the motor acts as an ideal integrator.

Effect of load on servomotor dynamics. Most important among the characteristics of the servomotor is the maximum acceleration obtainable. For a given available torque, the rotor moment of inertia must be a minimum. Since the servomotor operates under continuously varying conditions, acceleration and deceleration of the rotor occur from time to time. The servomotor must be able to absorb mechanical energy as well as to generate it. The performance of the servomotor when used as a brake should be satisfactory.

Let J_m and b_m be, respectively, the moment of inertia and viscous-friction coefficient of the rotor, and let J_L and b_L be, respectively, the moment of inertia and viscous-friction coefficient of the load on the output shaft. Assume that the moment of inertia and viscous-friction coefficient of the gear train are either negligible or included in J_L and b_L, respectively. Then, the equivalent moment of inertia J_{eq} referred to the motor shaft and equivalent viscous-friction coefficient b_{eq} referred to the motor shaft can be written as (for details, refer to Problem A-4-4)

$$J_{eq} = J_m + n^2 J_L$$

$$b_{eq} = b_m + n^2 f_L$$

where $n(n < 1)$ is the gear ratio between the motor and load. If the gear ratio n is small and $J_m \gg n^2 J_L$, then the moment of inertia of the load referred to the motor shaft is negligible with respect to the rotor moment of inertia. A similar argument applies to the load friction. In general, when the gear ratio n is small, the transfer function of the electric servomotor may be obtained without taking into account the load moment of inertia and friction. If neither J_m nor $n^2 J_L$ is negligibly small compared with the other, however, then the equivalent moment of inertia J_{eq} must be used for evaluating the transfer function of the motor–load combination.

Step response of second-order systems. The closed-loop transfer function of the system shown in Figure 4–8(c) is

$$\frac{C(s)}{R(s)} = \frac{K}{Js^2 + Bs + K} \tag{4–17}$$

which can be rewritten as

$$\frac{C(s)}{R(s)} = \frac{\frac{K}{J}}{\left[s + \frac{B}{2J} + \sqrt{\left(\frac{B}{2J}\right)^2 - \frac{K}{J}}\right]\left[s + \frac{B}{2J} - \sqrt{\left(\frac{B}{2J}\right)^2 - \frac{K}{J}}\right]} \tag{4–18}$$

The closed-loop poles are complex if $B^2 - 4JK < 0$, and they are real if $B^2 - 4JK \geq 0$. In transient-response analysis, it is convenient to write

$$\frac{K}{J} = \omega_n^2, \qquad \frac{B}{J} = 2\zeta\omega_n = 2\sigma$$

where σ is called the *attenuation;* ω_n, the *undamped natural frequency;* and ζ, the *damping ratio* of the system. The damping ratio ζ is the ratio of the actual damping B to the critical damping $B_c = 2\sqrt{JK}$ or

$$\zeta = \frac{B}{B_c} = \frac{B}{2\sqrt{JK}}$$

In terms of ζ and ω_n, the system shown in Figure 4–8(c) can be modified to that shown in Figure 4–9, and the closed-loop transfer function $C(s)/R(s)$ given by Equation (4–18) can be written

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{4–19}$$

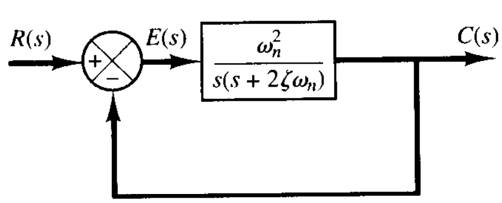


Figure 4–9
Second-order system.

The dynamic behavior of the second-order system can then be described in terms of two parameters ζ and ω_n. If $0 < \zeta < 1$, the closed-loop poles are complex conjugates and lie in the left-half s plane. The system is then called underdamped, and the transient response is oscillatory. If $\zeta = 1$, the system is called critically damped. Overdamped systems correspond to $\zeta > 1$. The transient response of critically damped and overdamped systems do not oscillate. If $\zeta = 0$, the transient response does not die out.

We shall now solve for the response of the system shown in Figure 4–9 to a unit-step input. We shall consider three different cases: the underdamped ($0 < \zeta < 1$), critically damped ($\zeta = 1$), and overdamped ($\zeta > 1$) cases.

<u>(1) Underdamped case ($0 < \zeta < 1$)</u>: In this case, $C(s)/R(s)$ can be written

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s + \zeta\omega_n + j\omega_d)(s + \zeta\omega_n - j\omega_d)}$$

where $\omega_d = \omega_n\sqrt{1 - \zeta^2}$. The frequency ω_d is called the *damped natural frequency*. For a unit-step input, $C(s)$ can be written

$$C(s) = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)s} \tag{4–20}$$

The inverse Laplace transform of Equation (4–20) can be obtained easily if $C(s)$ is written in the following form:

$$\begin{aligned} C(s) &= \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} \\ &= \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_d^2} - \frac{\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_d^2} \end{aligned}$$

In Chapter 2 it was shown that

$$\mathcal{L}^{-1}\left[\frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_d^2}\right] = e^{-\zeta\omega_n t}\cos\omega_d t$$

$$\mathcal{L}^{-1}\left[\frac{\omega_d}{(s + \zeta\omega_n)^2 + \omega_d^2}\right] = e^{-\zeta\omega_n t}\sin\omega_d t$$

Hence the inverse Laplace transform of Equation (4–20) is obtained as

$$\begin{aligned} \mathcal{L}^{-1}[C(s)] &= c(t) \\ &= 1 - e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1 - \zeta^2}}\sin\omega_d t\right) \\ &= 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}}\sin\left(\omega_d t + \tan^{-1}\frac{\sqrt{1 - \zeta^2}}{\zeta}\right), \qquad \text{for } t \geq 0 \end{aligned} \tag{4–21}$$

This result can be obtained directly by using a table of Laplace transforms. From Equation (4–21), it can be seen that the frequency of transient oscillation is the damped natural frequency ω_d and thus varies with the damping ratio ζ. The error signal for this system is the difference between the input and output and is

$$
\begin{aligned}
e(t) &= r(t) - c(t) \\
&= e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right) \qquad \text{for } t \geq 0
\end{aligned}
$$

This error signal exhibits a damped sinusoidal oscillation. At steady state, or at $t = \infty$, no error exists between the input and output.

If the damping ratio ζ is equal to zero, the response becomes undamped and oscillations continue indefinitely. The response $c(t)$ for the zero damping case may be obtained by substituting $\zeta = 0$ in Equation (4–21), yielding

$$
c(t) = 1 - \cos\omega_n t, \qquad \text{for } t \geq 0 \tag{4–22}
$$

Thus, from Equation (4–22), we see that ω_n represents the undamped natural frequency of the system. That is, ω_n is that frequency at which the system would oscillate if the damping were decreased to zero. If the linear system has any amount of damping, the undamped natural frequency cannot be observed experimentally. The frequency that may be observed is the damped natural frequency ω_d, which is equal to $\omega_n\sqrt{1-\zeta^2}$. This frequency is always lower than the undamped natural frequency. An increase in ζ would reduce the damped natural frequency ω_d. If ζ is increased beyond unity, the response becomes overdamped and will not oscillate.

(2) Critically damped case ($\zeta = 1$): If the two poles of $C(s)/R(s)$ are nearly equal, the system may be approximated by a critically damped one.

For a unit-step input, $R(s) = 1/s$ and $C(s)$ can be written

$$
C(s) = \frac{\omega_n^2}{(s + \omega_n)^2 s} \tag{4–23}
$$

The inverse Laplace transform of Equation (4–23) may be found as

$$
c(t) = 1 - e^{-\omega_n t}(1 + \omega_n t), \qquad \text{for } t \geq 0 \tag{4–24}
$$

This result can be obtained by letting ζ approach unity in Equation (4–21) and by using the following limit:

$$
\lim_{\zeta\to 1}\frac{\sin\omega_d t}{\sqrt{1-\zeta^2}} = \lim_{\zeta\to 1}\frac{\sin\omega_n\sqrt{1-\zeta^2}\,t}{\sqrt{1-\zeta^2}} = \omega_n t
$$

(3) Overdamped case ($\zeta > 1$): In this case, the two poles of $C(s)/R(s)$ are negative real and unequal. For a unit-step input, $R(s) = 1/s$ and $C(s)$ can be written

$$
C(s) = \frac{\omega_n^2}{(s + \zeta\omega_n + \omega_n\sqrt{\zeta^2-1})(s + \zeta\omega_n - \omega_n\sqrt{\zeta^2-1})s} \tag{4–25}
$$

The inverse Laplace transform of Equation (4–25) is

$$
\begin{aligned}
c(t) = 1 &+ \frac{1}{2\sqrt{\zeta^2-1}(\zeta + \sqrt{\zeta^2-1})}e^{-(\zeta+\sqrt{\zeta^2-1})\omega_n t} \\
&- \frac{1}{2\sqrt{\zeta^2-1}(\zeta - \sqrt{\zeta^2-1})}e^{-(\zeta-\sqrt{\zeta^2-1})\omega_n t}
\end{aligned}
$$

$$= 1 + \frac{\omega_n}{2\sqrt{\zeta^2 - 1}}\left(\frac{e^{-s_1 t}}{s_1} - \frac{e^{-s_2 t}}{s_2}\right), \qquad \text{for } t \geq 0 \tag{4–26}$$

where $s_1 = (\zeta + \sqrt{\zeta^2 - 1})\omega_n$ and $s_2 = (\zeta - \sqrt{\zeta^2 - 1})\omega_n$. Thus, the response $c(t)$ includes two decaying exponential terms.

When ζ is appreciably greater than unity, one of the two decaying exponentials decreases much faster than the other, so the faster decaying exponential term (which corresponds to a smaller time constant) may be neglected. That is, if $-s_2$ is located very much closer to the $j\omega$ axis than $-s_1$ (which means $|s_2| \ll |s_1|$), then for an approximate solution we may neglect $-s_1$. This is permissible because the effect of $-s_1$ on the response is much smaller than that of $-s_2$, since the term involving s_1 in Equation (4–26) decays much faster than the term involving s_2. Once the faster decaying exponential term has disappeared, the response is similar to that of a first-order system, and $C(s)/R(s)$ may be approximated by

$$\frac{C(s)}{R(s)} = \frac{\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}}{s + \zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}} = \frac{s_2}{s + s_2}$$

This approximate form is a direct consequence of the fact that the initial values and final values of both the original $C(s)/R(s)$ and the approximate one agree with each other.

With the approximate transfer function $C(s)/R(s)$, the unit-step response can be obtained as

$$C(s) = \frac{\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}}{(s + \zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1})s}$$

The time response $c(t)$ is then

$$c(t) = 1 - e^{-(\zeta - \sqrt{\zeta^2 - 1})\omega_n t}, \qquad \text{for } t \geq 0$$

This gives an approximate unit-step response when one of the poles of $C(s)/R(s)$ can be neglected.

A family of curves $c(t)$ with various values of ζ is shown in Figure 4–10, where the abscissa is the dimensionless variable $\omega_n t$. The curves are functions only of ζ. These

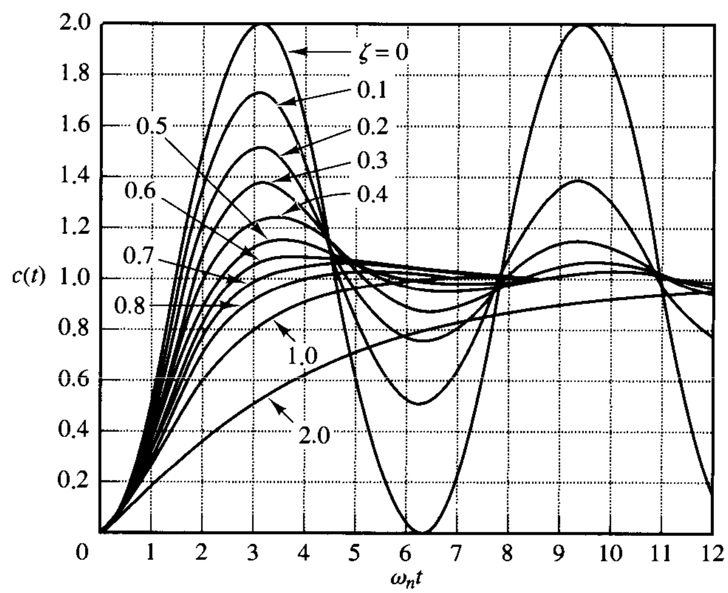


Figure 4–10
Unit-step response curves of the system shown in Figure 4–9.

curves are obtained from Equations (4–21), (4–24), and (4–26). The system described by these equations was initially at rest.

Note that two second-order systems having the same ζ but different ω_n will exhibit the same overshoot and the same oscillatory pattern. Such systems are said to have the same relative stability.

It is important to note that, for second-order systems whose closed-loop transfer functions are different from that given by Equation (4–19), the step-response curves may look quite different from those shown in Figure 4–10.

From Figure 4–10, we see that an underdamped system with ζ between 0.5 and 0.8 gets close to the final value more rapidly than a critically damped or overdamped system. Among the systems responding without oscillation, a critically damped system exhibits the fastest response. An overdamped system is always sluggish in responding to any inputs.

Definitions of transient-response specifications. In many practical cases, the desired performance characteristics of control systems are specified in terms of time-domain quantities. Systems with energy storage cannot respond instantaneously and will exhibit transient responses whenever they are subjected to inputs or disturbances.

Frequently, the performance characteristics of a control system are specified in terms of the transient response to a unit-step input since it is easy to generate and is sufficiently drastic. (If the response to a step input is known, it is mathematically possible to compute the response to any input.)

The transient response of a system to a unit-step input depends on the initial conditions. For convenience in comparing transient responses of various systems, it is a common practice to use the standard initial condition that the system is at rest initially with output and all time derivatives thereof zero. Then the response characteristics can be easily compared.

The transient response of a practical control system often exhibits damped oscillations before reaching steady state. In specifying the transient-response characteristics of a control system to a unit-step input, it is common to specify the following:

1. Delay time, t_d
2. Rise time, t_r
3. Peak time, t_p
4. Maximum overshoot, M_p
5. Settling time, t_s

These specifications are defined in what follows and are shown graphically in Figure 4–11.

1. Delay time, t_d: The delay time is the time required for the response to reach half the final value the very first time.
2. Rise time, t_r: The rise time is the time required for the response to rise from 10% to 90%, 5% to 95%, or 0% to 100% of its final value. For underdamped second-order systems, the 0% to 100% rise time is normally used. For overdamped systems, the 10% to 90% rise time is commonly used.

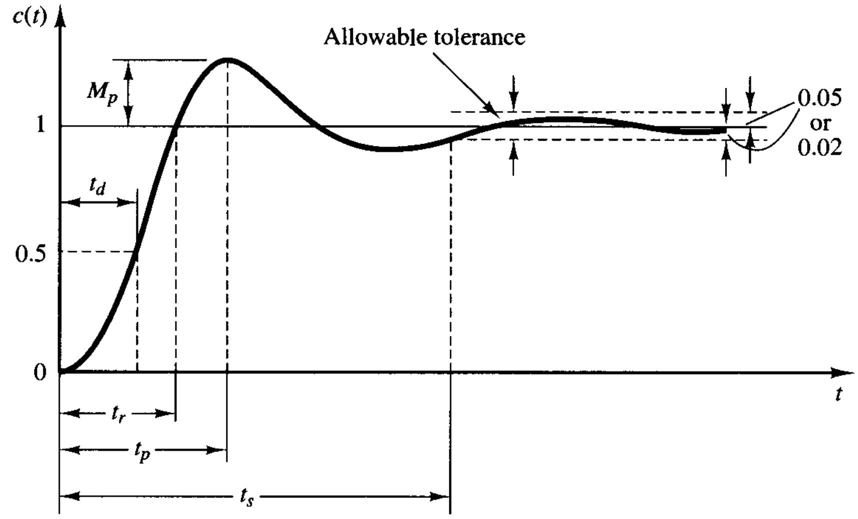


Figure 4–11
Unit-step response curve showing t_d, t_r, t_p, M_p, and t_s.

3. Peak time, t_p: The peak time is the time required for the response to reach the first peak of the overshoot.
4. Maximum (percent) overshoot, M_p: The maximum overshoot is the maximum peak value of the response curve measured from unity. If the final steady-state value of the response differs from unity, then it is common to use the maximum percent overshoot. It is defined by

$$\text{Maximum percent overshoot} = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100\%$$

   The amount of the maximum (percent) overshoot directly indicates the relative stability of the system.
5. Settling time, t_s: The settling time is the time required for the response curve to reach and stay within a range about the final value of size specified by absolute percentage of the final value (usually 2% or 5%). The settling time is related to the largest time constant of the control system. Which percentage error criterion to use may be determined from the objectives of the system design in question.

The time-domain specifications just given are quite important since most control systems are time-domain systems; that is, they must exhibit acceptable time responses. (This means that the control system must be modified until the transient response is satisfactory.) Note that if we specify the values of t_d, t_r, t_p, t_s, and M_p, then the shape of the response curve is virtually determined. This may be seen clearly from Figure 4–12.

Note that not all these specifications necessarily apply to any given case. For example, for an overdamped system, the terms peak time and maximum overshoot do not apply. (For systems that yield steady-state errors for step inputs, this error must be kept within a specified percentage level. Detailed discussions of steady-state errors are postponed until Section 5–10.)

A few comments on transient-response specifications. Except for certain applications where oscillations cannot be tolerated, it is desirable that the transient

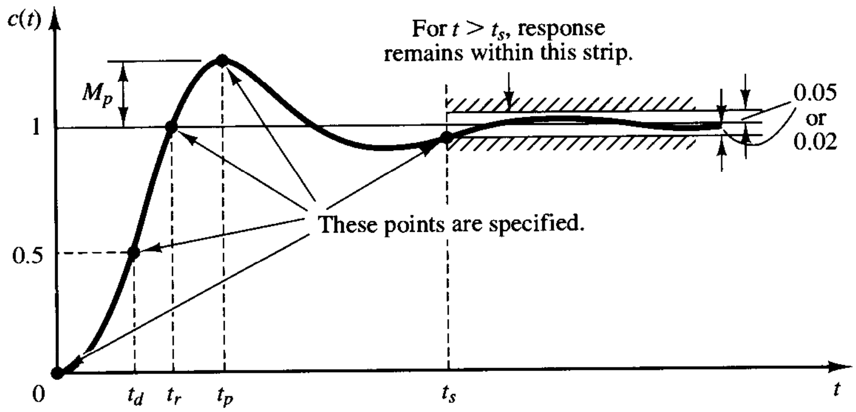


Figure 4–12 Transient-response specifications.

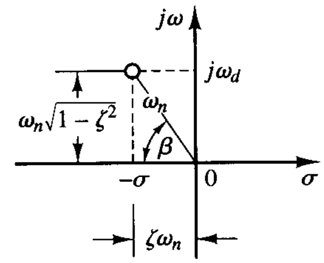


Figure 4–13 Definition of the angle β.

response be sufficiently fast and be sufficiently damped. Thus, for a desirable transient response of a second-order system, the damping ratio must be between 0.4 and 0.8. Small values of ζ ($\zeta < 0.4$) yield excessive overshoot in the transient response, and a system with a large value of ζ ($\zeta > 0.8$) responds sluggishly.

We shall see later that the maximum overshoot and the rise time conflict with each other. In other words, both the maximum overshoot and the rise time cannot be made smaller simultaneously. If one of them is made smaller, the other necessarily becomes larger.

Second-order systems and transient-response specifications. In the following, we shall obtain the rise time, peak time, maximum overshoot, and settling time of the second-order system given by Equation (4–19). These values will be obtained in terms of ζ and ω_n. The system is assumed to be underdamped.

Rise time t_r: Referring to Equation (4–21), we obtain the rise time t_r by letting $c(t_r) = 1$ or

$$c(t_r) = 1 = 1 - e^{-\zeta\omega_n t_r}\left(\cos \omega_d t_r + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t_r\right) \tag{4–27}$$

Since $e^{-\zeta\omega_n t_r} \neq 0$, we obtain from Equation (4–27) the following equation:

$$\cos \omega_d t_r + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t_r = 0$$

or

$$\tan \omega_d t_r = -\frac{\sqrt{1-\zeta^2}}{\zeta} = -\frac{\omega_d}{\sigma}$$

Thus, the rise time t_r is

$$t_r = \frac{1}{\omega_d} \tan^{-1}\left(\frac{\omega_d}{-\sigma}\right) = \frac{\pi - \beta}{\omega_d} \tag{4–28}$$

where β is defined in Figure 4–13. Clearly, for a small value of t_r, ω_d must be large.

Peak time t_p: Referring to Equation (4–21), we may obtain the peak time by differentiating $c(t)$ with respect to time and letting this derivative equal zero. Since

$$\frac{dc}{dt} = \zeta\omega_n e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right)$$
$$+ e^{-\zeta\omega_n t}\left(\omega_d \sin\omega_d t - \frac{\zeta\omega_d}{\sqrt{1-\zeta^2}}\cos\omega_d t\right)$$

and the cosine terms in this last equation cancel each other, dc/dt, evaluated at $t = t_p$, can be simplified to

$$\left.\frac{dc}{dt}\right|_{t=t_p} = (\sin\omega_d t_p)\frac{\omega_n}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t_p} = 0$$

This last equation yields the following equation:

$$\sin\omega_d t_p = 0$$

or

$$\omega_d t_p = 0, \pi, 2\pi, 3\pi, \ldots$$

Since the peak time corresponds to the first peak overshoot, $\omega_d t_p = \pi$. Hence

$$t_p = \frac{\pi}{\omega_d} \tag{4-29}$$

The peak time t_p corresponds to one-half cycle of the frequency of damped oscillation.

Maximum overshoot M_p: The maximum overshoot occurs at the peak time or at $t = t_p = \pi/\omega_d$. Thus, from Equation (4–21), M_p is obtained as

$$\begin{aligned} M_p &= c(t_p) - 1 \\ &= -e^{-\zeta\omega_n(\pi/\omega_d)}\left(\cos\pi + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\pi\right) \\ &= e^{-(\sigma/\omega_d)\pi} = e^{-(\zeta/\sqrt{1-\zeta^2})\pi} \end{aligned} \tag{4-30}$$

The maximum percent overshoot is $e^{-(\sigma/\omega_d)\pi} \times 100\%$.

Settling time t_s: For an underdamped second-order system, the transient response is obtained from Equation (4–21) as

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_d t + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right), \qquad \text{for } t \geq 0$$

The curves $1 \pm (e^{-\zeta\omega_n t}/\sqrt{1-\zeta^2})$ are the envelope curves of the transient response for a unit-step input. The response curve $c(t)$ always remains within a pair of the envelope curves, as shown in Figure 4–14. The time constant of these envelope curves is $1/\zeta\omega_n$.

The speed of decay of the transient response depends on the value of the time constant $1/\zeta\omega_n$. For a given ω_n, the settling time t_s is a function of the damping ratio ζ. From Figure 4–10, we see that for the same ω_n and for a range of ζ between 0 and 1 the settling time t_s for a very lightly damped system is larger than that for a properly damped system. For an overdamped system, the settling time t_s becomes large because of the sluggish start of the response.

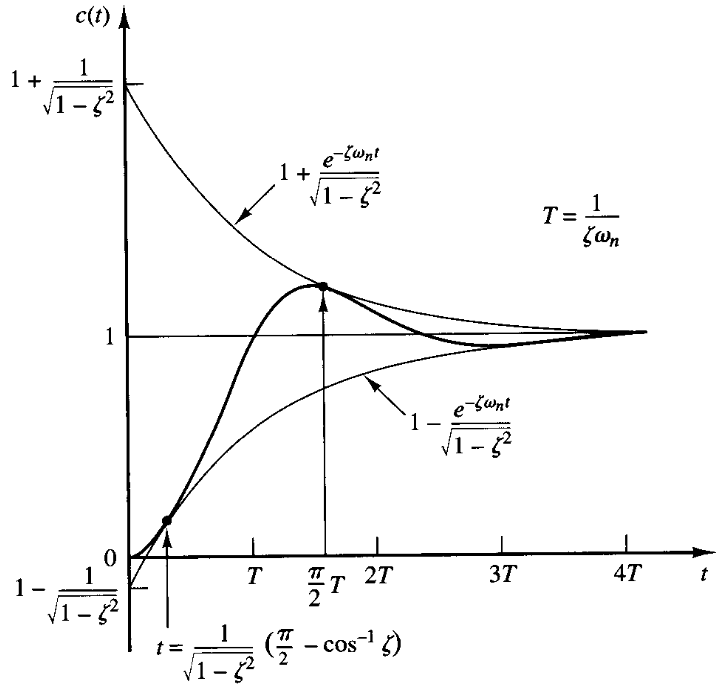


Figure 4–14
Pair of envelope curves for the unit-step response curve of the system shown in Figure 4–9.

The settling time corresponding to a ±2% or ±5% tolerance band may be measured in terms of the time constant $T = 1/\zeta\omega_n$ from the curves of Figure 4–10 for different values of ζ. The results are shown in Figure 4–15. For $0 < \zeta < 0.9$, if the 2% criterion is used t_s is approximately four times the time constant of the system. If the 5% criterion is used, then t_s is approximately three times the time constant. Note that the settling time reaches a minimum value around $\zeta = 0.76$ (for the 2% criterion) or $\zeta = 0.68$ (for the 5% criterion) and then increases almost linearly for large values of ζ. The discontinuities in the curves of Figure 4–15 arise because an infinitesimal change in the value of ζ can cause a finite change in the settling time.

For convenience in comparing the responses of systems, we commonly define the settling time t_s to be

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} \qquad \text{(2\% criterion)} \tag{4–31}$$

or

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} \qquad \text{(5\% criterion)} \tag{4–32}$$

Note that the settling time is inversely proportional to the product of the damping ratio and the undamped natural frequency of the system. Since the value of ζ is usually determined from the requirement of permissible maximum overshoot, the settling time is determined primarily by the undamped natural frequency ω_n. This means that the duration of the transient period may be varied, without changing the maximum overshoot, by adjusting the undamped natural frequency ω_n.

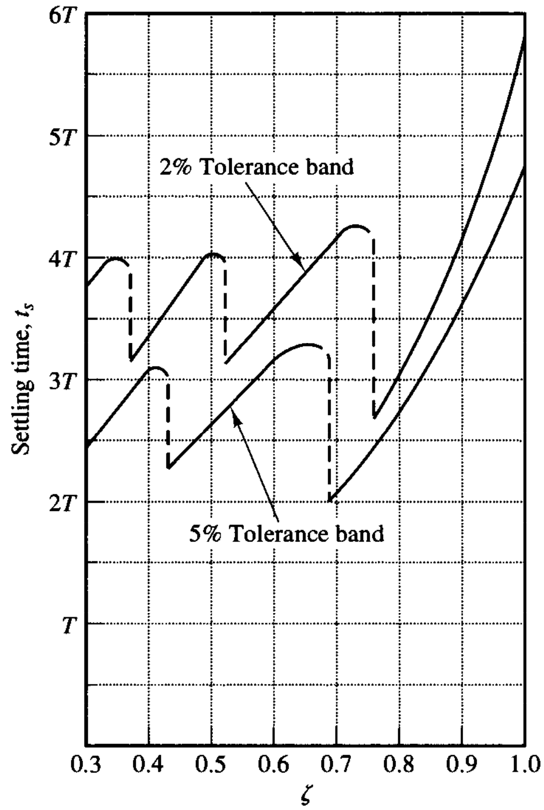


Figure 4–15
Settling time t_s versus ζ curves.

From the preceding analysis, it is evident that for rapid response ω_n must be large. To limit the maximum overshoot M_p and to make the settling time small, the damping ratio ζ should not be too small. The relationship between the maximum percent overshoot M_p and the damping ratio ζ is presented in Figure 4–16. Note that if the damping ratio is between 0.4 and 0.8 then the maximum percent overshoot for step response is between 25% and 2.5%.

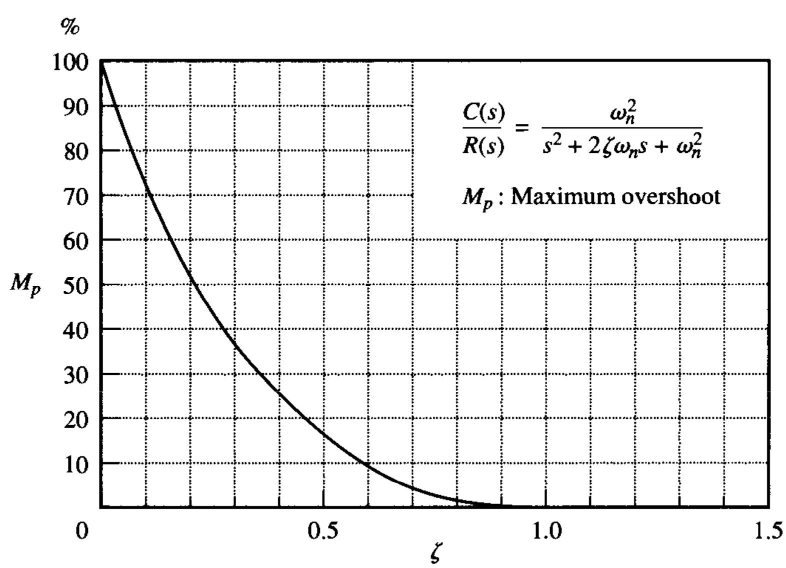


Figure 4–16
M_p versus ζ curve.

EXAMPLE 4–2 Consider the system shown in Figure 4–9, where $\zeta = 0.6$ and $\omega_n = 5$ rad/sec. Let us obtain the rise time t_r, peak time t_p, maximum overshoot M_p, and settling time t_s when the system is subjected to a unit-step input.

From the given values of ζ and ω_n, we obtain $\omega_d = \omega_n\sqrt{1 - \zeta^2} = 4$ and $\sigma = \zeta\omega_n = 3$.

Rise time t_r: The rise time is

$$t_r = \frac{\pi - \beta}{\omega_d} = \frac{3.14 - \beta}{4}$$

where β is given by

$$\beta = \tan^{-1}\frac{\omega_d}{\sigma} = \tan^{-1}\frac{4}{3} = 0.93 \text{ rad}$$

The rise time t_r is thus

$$t_r = \frac{3.14 - 0.93}{4} = 0.55 \text{ sec}$$

Peak time t_p: The peak time is

$$t_p = \frac{\pi}{\omega_d} = \frac{3.14}{4} = 0.785 \text{ sec}$$

Maximum overshoot M_p: The maximum overshoot is

$$M_p = e^{-(\sigma/\omega_d)\pi} = e^{-(3/4)\times 3.14} = 0.095$$

The maximum percent overshoot is thus 9.5%

Settling time t_s: For the 2% criterion, the settling time is

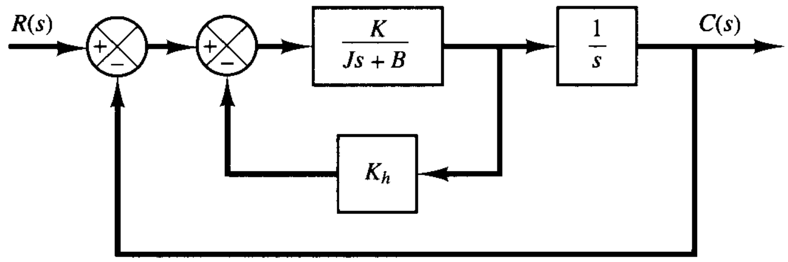
$$t_s = \frac{4}{\sigma} = \frac{4}{3} = 1.33 \text{ sec}$$

For the 5% criterion,

$$t_s = \frac{3}{\sigma} = \frac{3}{3} = 1 \text{ sec}$$

Servo system with velocity feedback. The derivative of the output signal can be used to improve system performance. In obtaining the derivative of the output position signal, it is desirable to use a tachometer instead of physically differentiating the output signal. (Note that the differentiation amplifies noise effects. In fact, if discontinuous noises are present, differentiation amplifies the discontinuous noises more than the useful signal. For example, the output of a potentiometer is a discontinuous voltage signal because, as the potentiometer brush is moving on the windings, voltages are induced in the switchover turns and thus generate transients. The output of the potentiometer therefore should not be followed by a differentiating element.)

Consider the servo system shown in Figure 4–17(a). In this device, the velocity signal, together with the positional signal, is fed back to the input to produce the actuating error signal. In any servo system, such a velocity signal can be easily generated by a



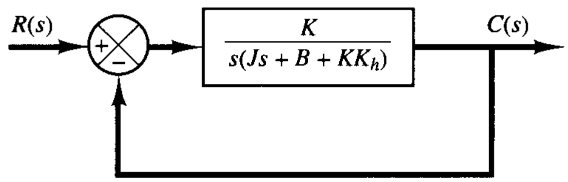


Figure 4–17
(a) Block diagram of a servo system; (b) simplified block diagram.

tachometer. The block diagram shown in Figure 4–17(a) can be simplified, as shown in Figure 4–17(b), giving

$$\frac{C(s)}{R(s)} = \frac{K}{Js^2 + (B + KK_h)s + K} \tag{4–33}$$

Comparing Equation (4–33) with Equation (4–17), notice that the velocity feedback has the effect of increasing damping. The damping ratio ζ becomes

$$\zeta = \frac{B + KK_h}{2\sqrt{KJ}} \tag{4–34}$$

The undamped natural frequency $\omega_n = \sqrt{K/J}$ is not affected by velocity feedback. Noting that the maximum overshoot for a unit-step input can be controlled by controlling the value of the damping ratio ζ, we can reduce the maximum overshoot by adjusting the velocity feedback constant K_h so that ζ is between 0.4 and 0.7.

Remember that velocity feedback has the effect of increasing the damping ratio without affecting the undamped natural frequency of the system.

EXAMPLE 4–3

For the system shown in Figure 4–17(a), determine the values of gain K and velocity feedback constant K_h so that the maximum overshoot in the unit-step response is 0.2 and the peak time is 1 sec. With these values of K and K_h, obtain the rise time and settling time. Assume that $J = 1$ kg-m^2 and $B = 1$ N-m/rad/sec.

Determination of the values of K and K_h: The maximum overshoot M_p is given by Equation (4–30) as

$$M_p = e^{-(\zeta/\sqrt{1-\zeta^2})\pi}$$

This value must be 0.2. Thus,

$$e^{-(\zeta/\sqrt{1-\zeta^2})\pi} = 0.2$$

or

$$\frac{\zeta\pi}{\sqrt{1 - \zeta^2}} = 1.61$$

which yields

$$\zeta = 0.456$$

The peak time t_p is specified as 1 sec; therefore, from Equation (4–29),

$$t_p = \frac{\pi}{\omega_d} = 1$$

or

$$\omega_d = 3.14$$

Since ζ is 0.456, ω_n is

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} = 3.53$$

Since the natural frequency ω_n is equal to $\sqrt{K/J}$,

$$K = J\omega_n^2 = \omega_n^2 = 12.5 \text{ N-m}$$

Then, K_h is, from Equation (4–34),

$$K_h = \frac{2\sqrt{KJ}\zeta - B}{K} = \frac{2\sqrt{K}\zeta - 1}{K} = 0.178 \text{ sec}$$

Rise time t_r: From Equation (4–28), the rise time t_r is

$$t_r = \frac{\pi - \beta}{\omega_d}$$

where

$$\beta = \tan^{-1}\frac{\omega_d}{\sigma} = \tan^{-1} 1.95 = 1.10$$

Thus, t_r is

$$t_r = 0.65 \text{ sec}$$

Settling time t_s: For the 2% criterion,

$$t_s = \frac{4}{\sigma} = 2.48 \text{ sec}$$

For the 5% criterion,

$$t_s = \frac{3}{\sigma} = 1.86 \text{ sec}$$

Impulse response of second-order systems. For a unit-impulse input $r(t)$, the corresponding Laplace transform is unity, or $R(s) = 1$. The unit-impulse response $C(s)$ of the second-order system shown in Figure 4–9 is

$$C(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The inverse Laplace transform of this equation yields the time solution for the response $c(t)$ as follows:

For $0 \le \zeta < 1$,

$$c(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_n \sqrt{1-\zeta^2}\, t, \qquad \text{for } t \ge 0 \tag{4–35}$$

For $\zeta = 1$,

$$c(t) = \omega_n^2 t e^{-\omega_n t}, \qquad \text{for } t \ge 0 \tag{4–36}$$

For $\zeta > 1$,

$$c(t) = \frac{\omega_n}{2\sqrt{\zeta^2-1}} e^{-(\zeta-\sqrt{\zeta^2-1})\omega_n t} - \frac{\omega_n}{2\sqrt{\zeta^2-1}} e^{-(\zeta+\sqrt{\zeta^2-1})\omega_n t}, \qquad \text{for } t \ge 0 \tag{4–37}$$

Note that without taking the inverse Laplace transform of $C(s)$ we can also obtain the time response $c(t)$ by differentiating the corresponding unit-step response since the unit-impulse function is the time derivative of the unit-step function. A family of unit-impulse response curves given by Equations (4–35) and (4–36) with various values of ζ is shown in Figure 4–18. The curves $c(t)/\omega_n$ are plotted against the dimensionless variable $\omega_n t$, and thus they are functions only of ζ. For the critically damped and overdamped cases, the unit-impulse response is always positive or zero; that is, $c(t) \ge 0$. This can be seen from Equations (4–36) and (4–37). For the underdamped case, the unit-impulse response $c(t)$ oscillates about zero and takes both positive and negative values.

From the foregoing analysis, we may conclude that if the impulse response $c(t)$ does not change sign, the system is either critically damped or overdamped, in which case the

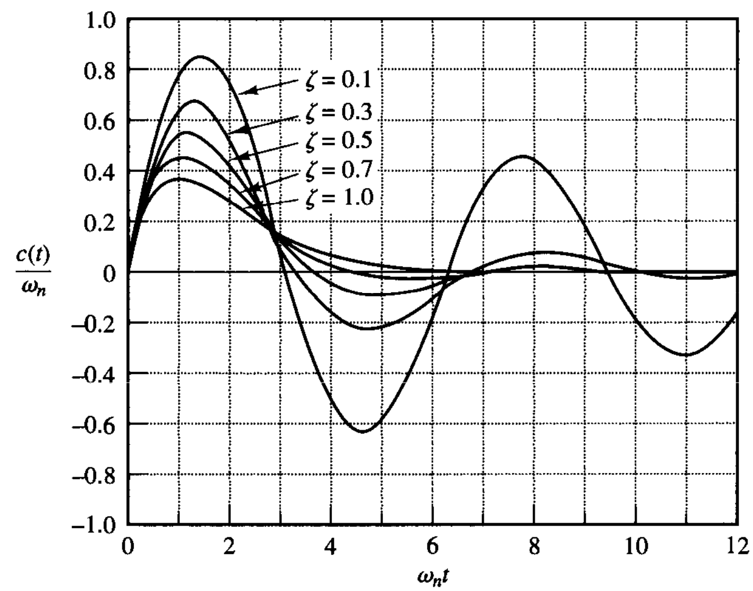


Figure 4–18 Unit-impulse response curves of the system shown in Figure 4–9.

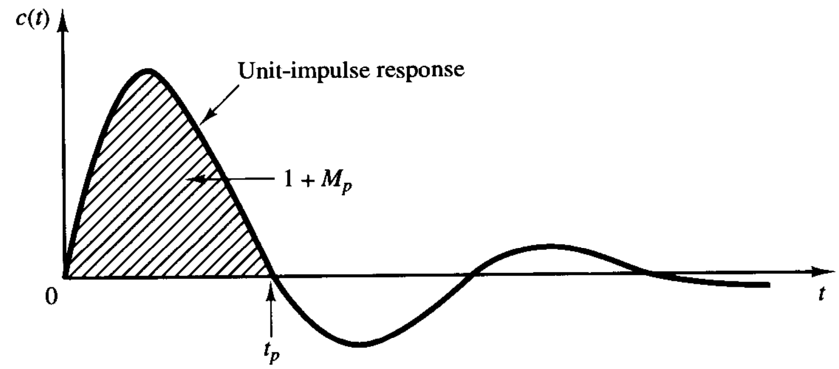


Figure 4–19 Unit-impulse response curve of the system shown in Figure 4–9.

corresponding step response does not overshoot but increases or decreases monotonically and approaches a constant value.

The maximum overshoot for the unit-impulse response of the underdamped system occurs at

$$t = \frac{\tan^{-1}\dfrac{\sqrt{1-\zeta^2}}{\zeta}}{\omega_n\sqrt{1-\zeta^2}}, \qquad \text{where } 0 < \zeta < 1$$

and the maximum overshoot is

$$c(t)_{\max} = \omega_n \exp\left(-\frac{\zeta}{\sqrt{1-\zeta^2}}\tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right), \qquad \text{where } 0 < \zeta < 1$$

Since the unit-impulse response function is the time derivative of the unit-step response function, the maximum overshoot M_p for the unit-step response can be found from the corresponding unit-impulse response. That is, the area under the unit-impulse response curve from $t = 0$ to the time of the first zero, as shown in Figure 4–19, is $1 + M_p$, where M_p is the maximum overshoot (for the unit-step response) given by Equation (4–30). The peak time t_p (for the unit-step response) given by Equation (4–29) corresponds to the time that the unit-impulse response first crosses the time axis.

4–4 TRANSIENT-RESPONSE ANALYSIS WITH MATLAB

Introduction. In this section we present the computational approach to the transient-response analysis with MATLAB. Those readers who are as yet unfamiliar with MATLAB may wish to read Appendix before studying this section.

As stated earlier in this chapter, transient responses (such as the step response, impulse response, and ramp response) are used frequently to investigate the time-domain characteristics of control systems.

MATLAB representation of linear systems. The transfer function of a system is represented by two arrays of numbers. Consider the system

$$\frac{C(s)}{R(s)} = \frac{25}{s^2 + 4s + 25} \tag{4–38}$$

This system is represented as two arrays each containing the coefficients of the polynomials in decreasing powers of s as follows:

```
num = [0  0  25]
den = [1  4  25]
```

Note that zeros are padded where necessary.

If num and den (the numerator and denominator of the closed-loop transfer function) are known, commands such as

```
step(num,den),      step(num,den,t)
```

will generate plots of unit-step responses. (t in the step command is the user-specified time.)

For a control system defined in a state-space form, where state matrix $\mathbf{A}$, control matrix $\mathbf{B}$, output matrix $\mathbf{C}$, and direct transmission matrix $\mathbf{D}$ of state-space equations are known, the command

```
step(A,B,C,D)
```

will generate plots of unit-step responses. The time vector is automatically determined when t is not explicitly included in the step commands.

Note that when step commands have left-hand arguments such as

```
[y,x,t] = step(num,den,t)
[y,x,t] = step(A,B,C,D,iu)
[y,x,t] = step(A,B,C,D,iu,t)
```
(4–39)

no plot is shown on the screen. Hence it is necessary to use a *plot* command to see the response curves. The matrices y and x contain the output and state response of the system, respectively, evaluated at the computation time points t. (y has as many columns as outputs and one row for each element in t. x has as many columns as states and one row for each element in t.)

Note in Equation (4–39) that the scalar iu is an index into the inputs of the system and specifies which input is to be used for the response, and t is the user-specified time. If the system involves multiple inputs and multiple outputs, the step command, such as given by Equation (4–39), produces a series of step response plots, one for each input and output combination of

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

(For details, see Example 4–4.)

Obtaining the unit-step response of the transfer-function system. Let us consider the unit-step response of the system given by Equation (4–38). MATLAB Program 4–1 will yield a plot of the unit-step response of this system. A plot of the unit-step response curve is shown in Figure 4–20.

```
MATLAB Program 4-1

% ---------Unit-step response----------

% ***** Enter the numerator and denominator of the transfer
% function *****

num = [0   0   25];
den = [1   4   25];

% ***** Enter the following step-response command *****

step(num,den)

% ***** Enter grid and title of the plot *****

grid
title ('Unit-Step Response of G(s) = 25/(s^2 + 4s + 25)')
```

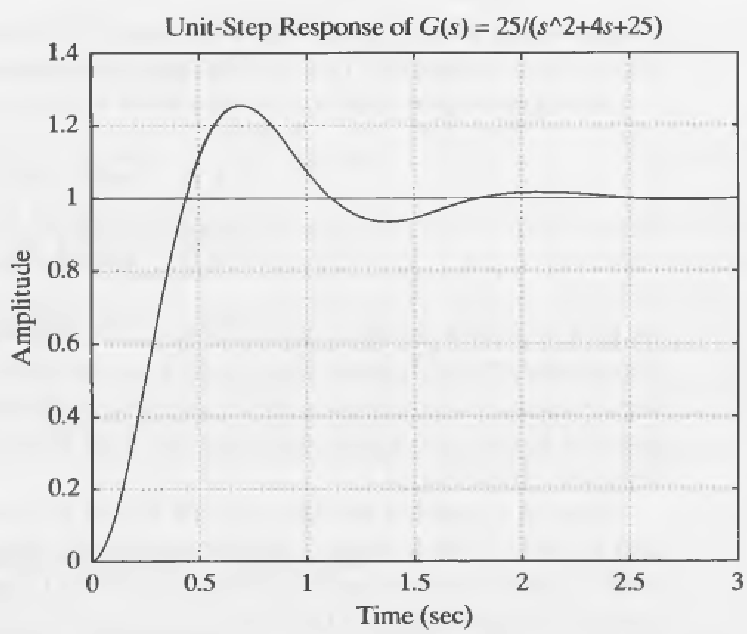


Figure 4–20
Unit-step response curve.

EXAMPLE 4–4 Consider the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 6.5 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

Obtain the unit-step response curves.

Although it is not necessary to obtain the transfer function expression for the system to obtain the unit-step response curves with MATLAB, we shall derive such an expression for reference. For the system defined by

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$$

$$\mathbf{y} = \mathbf{Cx} + \mathbf{Du}$$

the transfer matrix $\mathbf{G}(s)$ is a matrix that relates $\mathbf{Y}(s)$ and $\mathbf{U}(s)$ as follows:

$$\mathbf{Y}(s) = \mathbf{G}(s)\mathbf{U}(s)$$

Taking Laplace transforms of the state-space equations, we obtain

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{AX}(s) + \mathbf{BU}(s) \tag{4-40}$$

$$\mathbf{Y}(s) = \mathbf{CX}(s) + \mathbf{DU}(s) \tag{4-41}$$

In deriving the transfer matrix, we assume that $\mathbf{x}(0) = \mathbf{0}$. Then, from Equation (4–40), we get

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{BU}(s) \tag{4-42}$$

Substituting Equation (4–42) into Equation (4–41), we obtain

$$\mathbf{Y}(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}]\mathbf{U}(s)$$

Thus the transfer matrix $\mathbf{G}(s)$ is given by

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$

The transfer matrix $\mathbf{G}(s)$ for the given system becomes

$$\begin{aligned}
\mathbf{G}(s) &= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} s+1 & 1 \\ -6.5 & s \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \\
&= \frac{1}{s^2 + s + 6.5} \begin{bmatrix} s & -1 \\ 6.5 & s+1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \\
&= \frac{1}{s^2 + s + 6.5} \begin{bmatrix} s-1 & s \\ s+7.5 & 6.5 \end{bmatrix}
\end{aligned}$$

Hence

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} \dfrac{s-1}{s^2+s+6.5} & \dfrac{s}{s^2+s+6.5} \\ \dfrac{s+7.5}{s^2+s+6.5} & \dfrac{6.5}{s^2+s+6.5} \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix}$$

Since this system involves two inputs and two outputs, four transfer functions may be defined depending on which signals are considered as input and output. Note that, when considering the signal u_1 as the input, we assume that signal u_2 is zero, and vice versa. The four transfer functions are

$$\frac{Y_1(s)}{U_1(s)} = \frac{s-1}{s^2+s+6.5}, \qquad \frac{Y_2(s)}{U_1(s)} = \frac{s+7.5}{s^2+s+6.5}$$

$$\frac{Y_1(s)}{U_2(s)} = \frac{s}{s^2+s+6.5}, \qquad \frac{Y_2(s)}{U_2(s)} = \frac{6.5}{s^2+s+6.5}$$

The four individual step-response curves can be plotted by use of the command

```
step(A,B,C,D)
```

MATLAB Program 4–2 produces four such step-response curves. The curves are shown in Figure 4–21.

MATLAB Program 4–2
A = [−1 −1;6.5 0];   B = [1 1;1 0];   C = [1 0;0 1];   D = [0 0;0 0];   step(A,B,C,D)

To plot two step-response curves for the input u_1 in one diagram and two step-response curves for the input u_2 in another diagram, we may use the commands

```
step(A,B,C,D,1)
```

and

```
step(A,B,C,D,2)
```

respectively. MATLAB Program 4–3 is a program to plot two step-response curves for the input u_1 in one diagram and two step-response curves for the input u_2 in another diagram. Figure 4–22 shows the two diagrams, each consisting of two step-response curves.

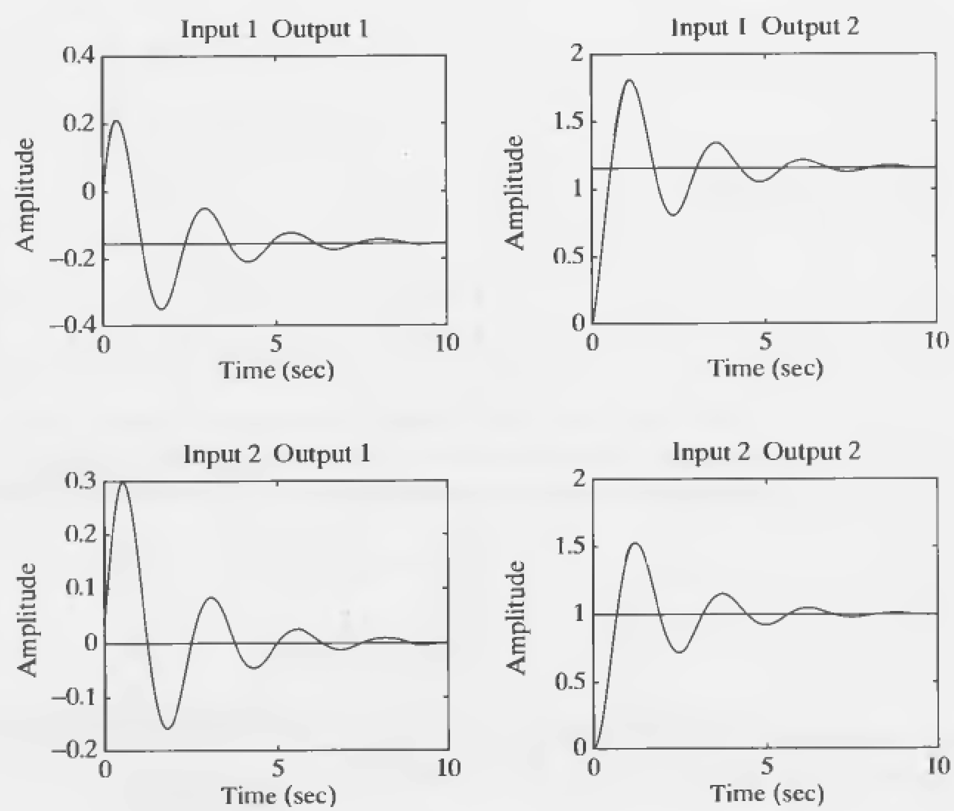


Figure 4–21
Unit-step response curves.

MATLAB Program 4–3

```
% ---------- Step-response curves for system defined in state
% space ----------

% ***** In this program we plot step-response curves of a system
% having two inputs (u1 and u2) and two outputs (y1 and y2) *****

% ***** We shall first plot step-response curves when the input is
% u1. Then we shall plot step-response curves when the input is
% u2 *****

% ***** Enter matrices A, B, C, and D *****

A = [-1   -1;6.5   0];
B = [1   1;1   0];
C = [1   0;0   1];
D = [0   0;0   0];

% ***** To plot step-response curves when the input is u1, enter
% the command 'step(A,B,C,D,1)' *****

step(A,B,C,D,1)
grid
title ('Step-Response Plots: Input = u1 (u2 = 0)')
text(3.4,-0.06,'Y1')
text(3.4,1.4,'Y2')

% ***** Next, we shall plot step-response curves when the input
% is u2. Enter the command 'step(A,B,C,D,2)' *****

step(A,B,C,D,2);
grid
title('Step-Response Plots: Input = u2 (u1 = 0)')
text(3,0.14,'Y1')
text(2.8,1.1,'Y2')
```

Writing text on the graphics screen. To write text on the graphics screen, enter, for example, the following statements:

```
text(3.4,-0.06,'Y1')
```

and

```
text(3.4,1.4,'Y2')
```

The first statement tells the computer to write 'Y1' beginning at the coordinates $x = 3.4$, $y = -0.06$. Similarly, the second statement tells the computer to write 'Y2' beginning at the coordinates $x = 3.4$, $y = 1.4$. [See MATLAB Program 4–3 and Figure 4–22(a).]

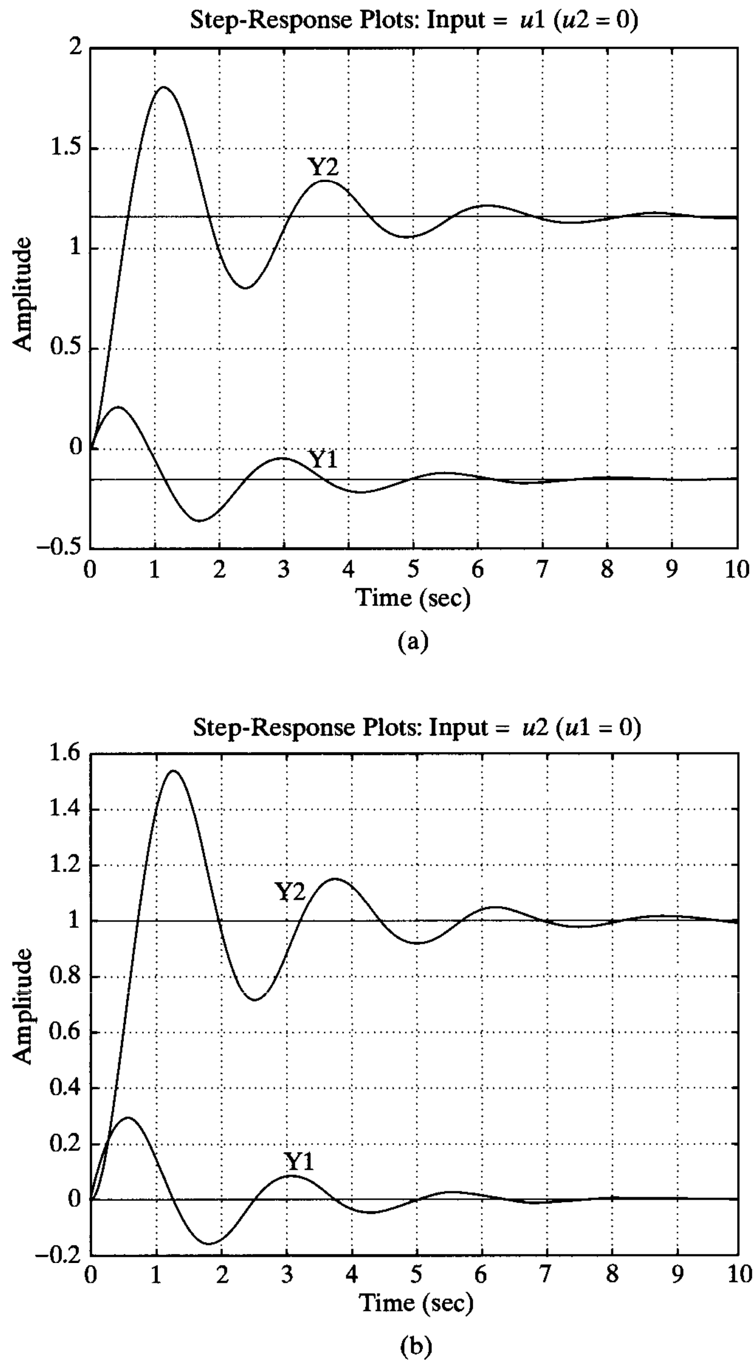


Figure 4–22 Unit-step response curves. (a) u_1 is the input ($u_2 = 0$); (b) u_2 is the input ($u_1 = 0$).

Impulse response. The unit-impulse response of a control system may be obtained by use of one of the following MATLAB commands:

```
impulse(num,den)
impulse(A,B,C,D)
[y,x,t] = impulse (num,den)
[y,x,t] = impulse(num,den,t)                    (4–43)
```

```
[y,x,t] = impulse(A,B,C,D)
[y,x,t] = impulse(A,B,C,D,iu)                    (4-44)
[y,x,t] = impulse (A,B,C,D,iu,t)                 (4-45)
```

The command "impulse(num,den)" plots the unit-impulse response on the screen. The command "impulse(A,B,C,D)" produces a series of unit-impulse-response plots, one for each input and output combination of the system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

with the time vector automatically determined. Note that in Equations (4–44) and (4–45) the scalar iu is an index into the inputs of the system and specifies which input to be used for the impulse response.

Note also that in Equations (4–43) and (4–45) t is the user-supplied time vector. The vector t specifies the times at which the impulse response is to be computed.

If MATLAB is invoked with the left-hand argument [y,x,t], such as in the case of [y,x,t] = impulse(A,B,C,D), the command returns the output and state responses of the system and the time vector t. No plot is drawn on the screen. The matrices y and x contain the output and state responses of the system evaluated at the time points t. (y has as many columns as outputs and one row for each element in t. x has as many columns as state variables and one row for each element in t.)

EXAMPLE 4–5 Obtain the unit-impulse response of the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0]u$$

A possible MATLAB program is shown in MATLAB Program 4–4. The resulting response curve is shown in Figure 4–23.

MATLAB Program 4–4

```
A = [0   1;-1   -1];
B = [0;1];
C = [1   0];
D = [0];
impulse(A,B,C,D);
grid;
title('Unit-Impulse Response')
```

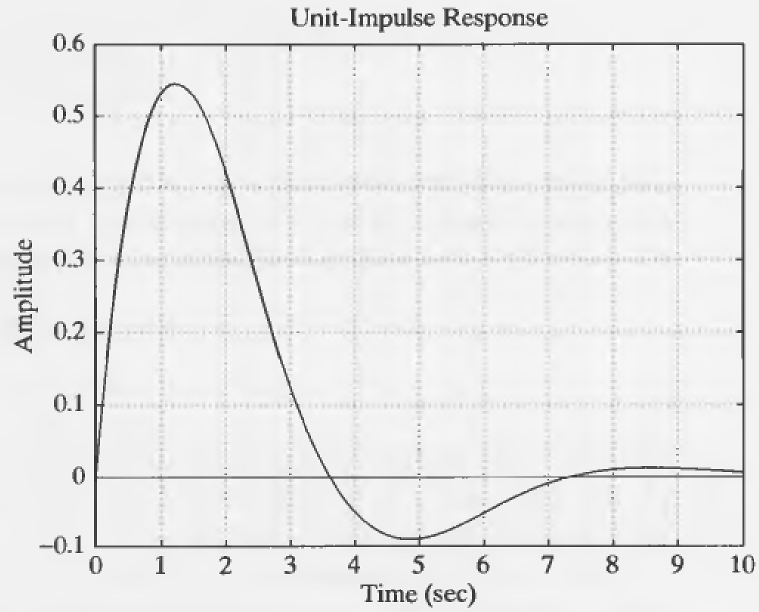


Figure 4–23 Unit-impulse response curve.

EXAMPLE 4–6 Obtain the unit-impulse response of the following system:

$$\frac{C(s)}{R(s)} = G(s) = \frac{1}{s^2 + 0.2s + 1}$$

MATLAB Program 4–5 will produce the unit-impulse response. The resulting plot is shown in Figure 4–24.

MATLAB Program 4–5

```
num = [0   0   1];
den = [1   0.2   1];
impulse(num,den);
grid
title('Unit-Impulse Response of G(s) = 1/(s^2 + 0.2s + l)')
```

Alternative approach to obtain impulse response. Note that when the initial conditions are zero the unit-impulse response of $G(s)$ is the same as the unit-step response of $sG(s)$.

Consider the unit-impulse response of the system considered in Example 4–6. Since $R(s) = 1$ for the unit-impulse input, we have

$$\frac{C(s)}{R(s)} = C(s) = G(s) = \frac{1}{s^2 + 0.2s + 1}$$

$$= \frac{s}{s^2 + 0.2s + 1}\frac{1}{s}$$

We can thus convert the unit-impulse response of $G(s)$ to the unit-step response of $sG(s)$.

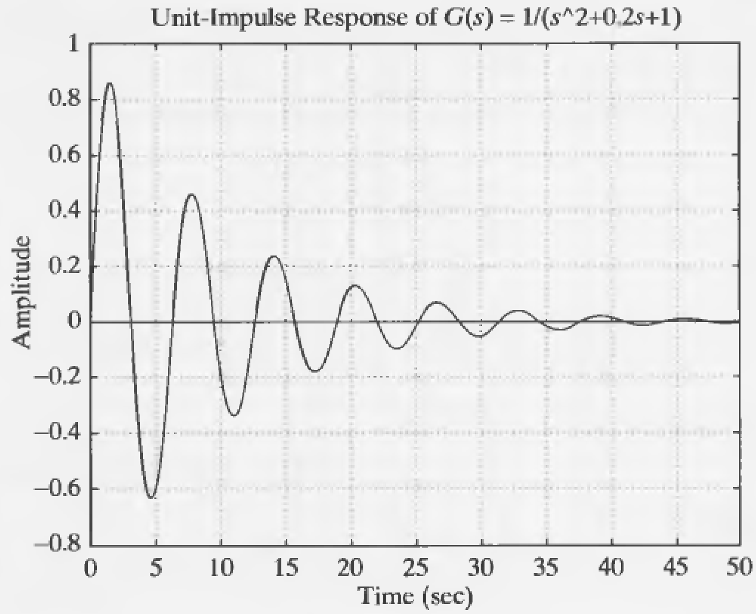


Figure 4–24 Unit-impulse response curve.

If we enter the following num and den into MATLAB,

```
num = [0   1   0]
den = [1   0.2   1]
```

and use the step-response command, as given in MATLAB Program 4–6, we obtain a plot of the unit-impulse response of the system as shown in Figure 4–25.

MATLAB Program 4–6

```
num = [0   1   0];
den = [1   0.2   1];
step(num,den);
grid;
title('Unit-Step Response of sG(s) = s/(s^2 + 0.2s + 1)')
```

Notice in Figure 4–25 (and many others) that the x axis and y axis labels are automatically determined. If it is desired to label the x axis and y axis differently, we need to modify the step command. For example, if it is desired to label the x axis as 't Sec' and the y axis as 'Input and Output,' then use step-response commands with left-hand arguments, such as

```
c = step(num,den,t)
```

or, more generally,

```
[y,x,t] = step(num,den,t)
```

See MATLAB Program 4–7.

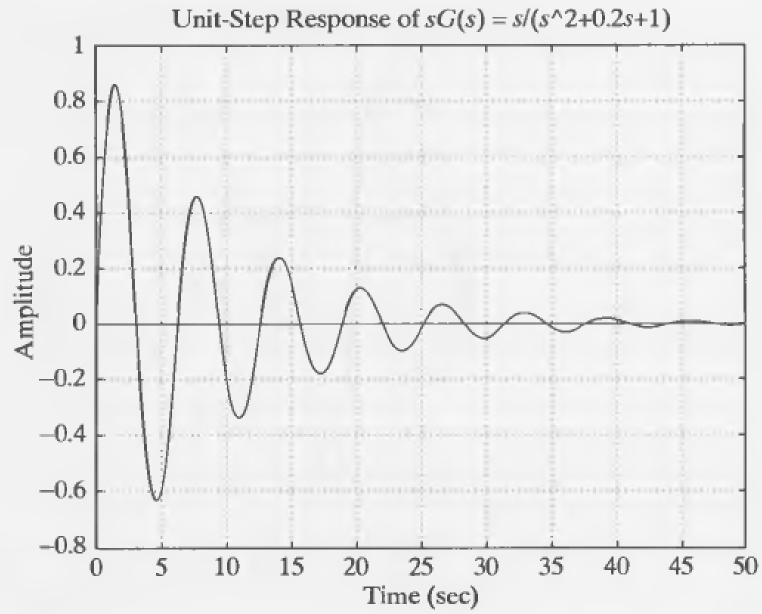


Figure 4–25 Unit-impulse response curve obtained as the unit-step response of $sG(s) = s/(s^2 + 0.2s + 1)$.

MATLAB Program 4–7

```
% ---------- Unit-ramp response ----------

% ***** The unit-ramp response is obtained as the unit-step
% response of G(s)/s *****

% ***** Enter the numerator and denominator of G(s)/s *****

num = [0   0   0   1];
den = [1   1   1   0];

% ***** Specify the computing time points (such as t = 0:0.1:7)
% and then enter step-response command: c = step(num,den,t) *****

t = 0:0.1:7;
c = step(num,den,t);

% ***** In plotting the ramp-response curve, add the reference
% input to the plot. The reference input is t. Add to the
% argument of the plot command with the following: t,t,'-'. Thus
% the plot command becomes as follows: plot(t,c,'o',t,t,'-') *****

plot(t,c,'o',t,t,'-')

% ***** Add grid, title, xlabel, and ylabel *****

grid
title('Unit-Ramp Response Curve for System G(s) = 1/(s^2 + s + 1)')
xlabel('t Sec')
ylabel('Input and Output')
```

Ramp response. There is no ramp command in MATLAB. Therefore, we need to use the step command to obtain the ramp response. Specifically, to obtain the ramp response of the transfer-function system $G(s)$, divide $G(s)$ by s and use the step-response command. For example, consider the closed-loop system

$$\frac{C(s)}{R(s)} = \frac{1}{s^2 + s + 1}$$

For a unit-ramp input, $R(s) = 1/(s^2)$. Hence

$$C(s) = \frac{1}{s^2 + s + 1}\frac{1}{s^2} = \frac{1}{(s^2 + s + 1)s}\frac{1}{s}$$

To obtain the unit-ramp response of this system, enter the following numerator and denominator into the MATLAB program,

```
num = [0  0  0  1];
den = [1  1  1  0];
```

and use the step-response command. See MATLAB Program 4–7. The plot obtained by using this program is shown in Figure 4–26.

Unit-ramp response of a system defined in state space. Next, we shall treat the unit-ramp response of the system in state-space form. Consider the system described by

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x} + Du$$

In what follows, we shall consider a simple example to explain the method. Let us assume that

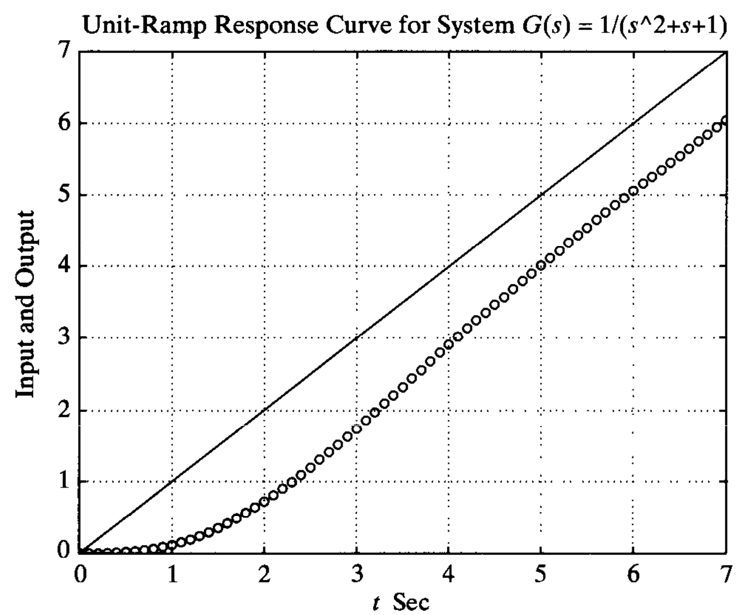


Figure 4–26 Unit-ramp response curve.

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{x}(0) = \mathbf{0}$$

$$\mathbf{C} = [1 \quad 0], \qquad D = [0]$$

When the initial conditions are zeros, the unit-ramp response is the integral of the unit-step response. Hence the unit-ramp response can be given by

$$z = \int_0^t y\,dt \tag{4-46}$$

From Equation (4–46), we obtain

$$\dot{z} = y = x_1 \tag{4-47}$$

Let us define

$$z = x_3$$

Then Equation (4–47) becomes

$$\dot{x}_3 = x_1 \tag{4-48}$$

Combining Equation (4–48) with the original state-space equation, we obtain

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$

$$z = [0 \quad 0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

which can be written as

$$\dot{\mathbf{x}} = \mathbf{AAx} + \mathbf{BB}u$$

$$z = \mathbf{CCx} + DDu$$

where

$$\mathbf{AA} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \left[\begin{array}{c|c} \mathbf{A} & \begin{matrix} 0 \\ 0 \end{matrix} \\ \hline \mathbf{C} & 0 \end{array}\right]$$

$$\mathbf{BB} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \left[\begin{array}{c} \mathbf{B} \\ \hline 0 \end{array}\right], \qquad \mathbf{CC} = [0 \quad 0 \quad 1], \qquad DD = [0]$$

Note that x_3 is the third element of $\mathbf{x}$. A plot of the unit-ramp response curve $z(t)$ can be obtained by entering MATLAB Program 4–8 into the computer. A plot of

```
MATLAB Program 4-8
% ---------- Unit-ramp response ----------

% ***** The unit-ramp response is obtained by adding a new
% state variable x3. The dimension of the state equation
% is enlarged by one *****

% ***** Enter matrices A, B, C, and D of the original state
% equation and output equation *****

A = [0   1;-1   -1];
B = [0;1];
C = [1   0];
D = [0];

% ***** Enter matrices AA, BB, CC, and DD of the new,
% enlarged state equation and output equation *****

AA = [A   zeros(2,1);C   0];
BB = [B;0];
CC = [0   0   1];
DD = [0];

% ***** Enter step-response command: [z,x,t] = step(AA,BB,CC,DD) *****

[z,x,t] = step(AA,BB,CC,DD);

% ***** In plotting x3 add the unit-ramp input t in the plot
% by entering the following command: plot(t,x3,'o',t,t,'-') *****

x3 = [0   0   1]*x'; plot(t,x3,'o',t,t,'-')
grid
title('Unit-Ramp Response')
xlabel('t Sec')
ylabel('Input and Output')
```

the unit-ramp response curve obtained from this MATLAB program is shown in Figure 4–27.

Response to initial condition (transfer-function approach). In what follows we shall present a method for obtaining the response to an initial condition by use of an example.

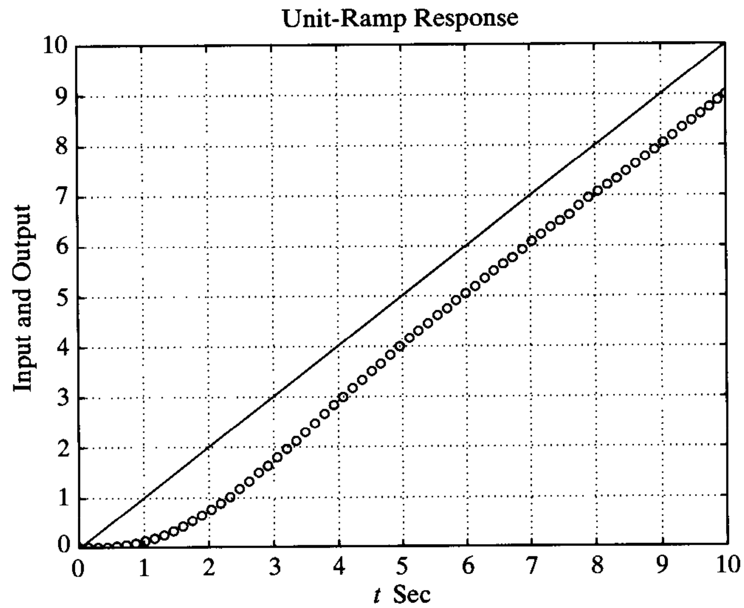


Figure 4–27 Unit-ramp response curve.

EXAMPLE 4–7 In this example, we shall consider a system subjected only to an initial condition.

Consider the mechanical system shown in Figure 4–28, where $m = 1$ kg, $b = 3$ N-sec/m, and $k = 2$ N/m. Assume that at $t = 0$ the mass m is pulled downward such that $x(0) = 0.1$ m and $\dot{x}(0) = 0.05$ m/sec. Obtain the motion of the mass subjected to the initial condition. (Assume no external forcing function.)

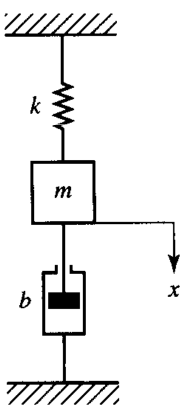


Figure 4–28 Mechanical system.

The system equation is

$$m\ddot{x} + b\dot{x} + kx = 0$$

with the initial conditions $x(0) = 0.1$ m and $\dot{x}(0) = 0.05$ m/sec. The Laplace transform of the system equation gives

$$m[s^2X(s) - sx(0) - \dot{x}(0)] + b[sX(s) - x(0)] + kX(s) = 0$$

or

$$(ms^2 + bs + k)X(s) = mx(0)s + m\dot{x}(0) + bx(0)$$

Solving this last equation for $X(s)$ and substituting the given numerical values, we obtain

$$X(s) = \frac{mx(0)s + m\dot{x}(0) + bx(0)}{ms^2 + bs + k}$$

$$= \frac{0.1s + 0.35}{s^2 + 3s + 2}$$

This equation can be written as

$$X(s) = \frac{0.1s^2 + 0.35s}{s^2 + 3s + 2}\frac{1}{s}$$

Hence the motion of the mass m may be obtained as the unit-step response of the following system:

$$G(s) = \frac{0.1s^2 + 0.35s}{s^2 + 3s + 2}$$

MATLAB Program 4–9 will give a plot of the motion of the mass. The plot is shown in Figure 4–29.

MATLAB Program 4–9

```
% ---------- Response to initial conditions ----------

% ***** System response to initial conditions is converted to
% a unit-step response by modifying the numerator polynomial *****

% ***** Enter the numerator and denominator of the transfer
% function G(s) *****

num = [0.1   0.35   0];
den = [1   3   2];

% ***** Enter the following step-response command *****

step(num,den)

% ***** Enter grid and title of the plot *****

grid
title('Response of Spring-Mass-Damper System to Initial Conditions')
```

Response to initial condition (state-space approach, case 1). Consider the system defined by

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \qquad \mathbf{x}(0) = \mathbf{x}_0 \tag{4–49}$$

Let us obtain the response $\mathbf{x}(t)$ when the initial condition $\mathbf{x}(0)$ is specified. (There is no external input function acting on this system.) Assume that $\mathbf{x}$ is an n-vector.

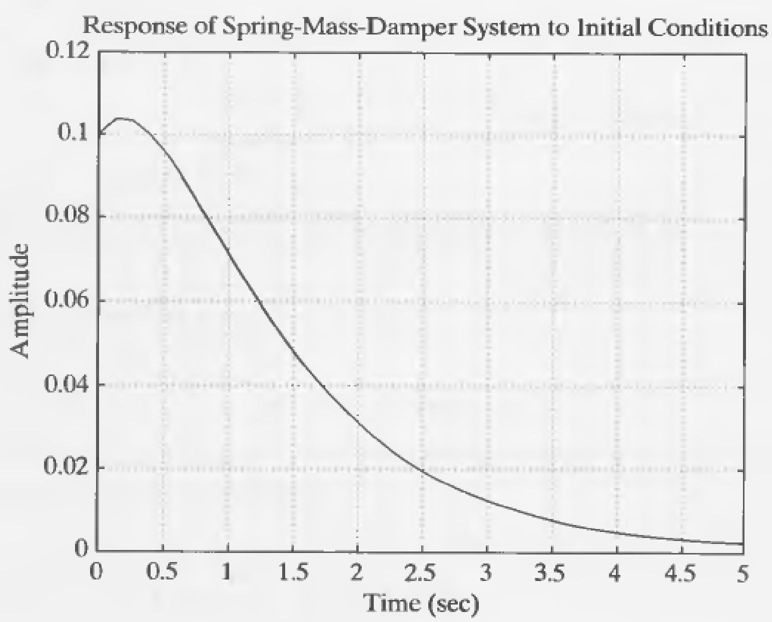


Figure 4–29 Response of the mechanical system considered in Example 4–7.

First, take Laplace transforms of both sides of Equation (4–49).

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s)$$

This equation can be rewritten as

$$s\mathbf{X}(s) = \mathbf{A}\mathbf{X}(s) + \mathbf{x}(0) \tag{4–50}$$

Taking the inverse Laplace transform of Equation (4–50), we obtain

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{x}(0)\,\delta(t) \tag{4–51}$$

(Notice that by taking the Laplace transform of a differential equation and then by taking the inverse Laplace transform of the Laplace-transformed equation we generate a differential equation that involves the initial condition.)

Now define

$$\dot{\mathbf{z}} = \mathbf{x} \tag{4–52}$$

Then Equation (4–51) can be written as

$$\ddot{\mathbf{z}} = \mathbf{A}\dot{\mathbf{z}} + \mathbf{x}(0)\,\delta(t) \tag{4–53}$$

By integrating Equation (4–53) with respect to t, we obtain

$$\dot{\mathbf{z}} = \mathbf{A}\mathbf{z} + \mathbf{x}(0)1(t) = \mathbf{A}\mathbf{z} + \mathbf{B}u \tag{4–54}$$

where

$$\mathbf{B} = \mathbf{x}(0), \qquad u = 1(t)$$

Referring to Equation (4–52), the state $\mathbf{x}(t)$ is given by $\dot{\mathbf{z}}(t)$. Thus,

$$\mathbf{x} = \dot{\mathbf{z}} = \mathbf{A}\mathbf{z} + \mathbf{B}u \tag{4–55}$$

Equation (4–55) gives the response to the initial condition.

Summarizing, the response of Equation (4–49) to the initial condition $\mathbf{x}(0)$ is obtained by solving the following state-space equations:

$$\dot{\mathbf{z}} = \mathbf{A}\mathbf{z} + \mathbf{B}u$$

$$\mathbf{x} = \mathbf{A}\mathbf{z} + \mathbf{B}u$$

where

$$\mathbf{B} = \mathbf{x}(0), \qquad u = 1(t)$$

MATLAB commands to obtain the response curves in one diagram are given next.

```
[x,z,t] = step(A,B,A,B);
x1 = [1  0  0 . . . 0]*x';
x2 = [0  1  0 . . . 0]*x';
   .
   .
   .
xn = [0  0  0 . . . 1]*x';
plot(t,xl,t,x2, . . . ,t,xn)
```

Response to initial condition (state-space approach, case 2). Consider the system defined by

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \qquad \mathbf{x}(0) = \mathbf{x}_0 \tag{4–56}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} \tag{4–57}$$

(Assume that $\mathbf{x}$ is an n-vector and $\mathbf{y}$ is an m-vector.)

Similar to case 1, by defining

$$\dot{\mathbf{z}} = \mathbf{x}$$

we can obtain the following equation:

$$\dot{\mathbf{z}} = \mathbf{A}\mathbf{z} + \mathbf{x}(0)1(t) = \mathbf{A}\mathbf{z} + \mathbf{B}u \tag{4–58}$$

where

$$\mathbf{B} = \mathbf{x}(0), \qquad u = 1(t)$$

Noting that $\mathbf{x} = \dot{\mathbf{z}}$, Equation (4–57) can be written as

$$\mathbf{y} = \mathbf{C}\dot{\mathbf{z}} \tag{4–59}$$

By substituting Equation (4–58) into Equation (4–59), we obtain

$$\mathbf{y} = \mathbf{C}(\mathbf{A}\mathbf{z} + \mathbf{B}u) = \mathbf{C}\mathbf{A}\mathbf{z} + \mathbf{C}\mathbf{B}u \tag{4–60}$$

The solution of Equations (4–58) and (4–60) gives the response of the system to a given initial condition. MATLAB commands to obtain the response curves (output curves y1 versus t, y2 versus t, . . . , ym versus t) are shown next.

```
[y,z,t] = step(A,B,C*A,C*B)
y1 = [1   0   0 . . . 0]*y';
y2 = [0   1   0 . . . 0]*y';
      .
      .
      .
ym = [0   0   0 . . . 1]*y';
plot(t,y1,t,y2 . . . ,t,ym)
```

EXAMPLE 4–8 Obtain the response of the system subjected to the given initial condition.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -10 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \qquad \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

or

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \qquad \mathbf{x}(0) = \mathbf{x}_0$$

Obtaining the response of the system to the given initial condition becomes that of solving the unit-step response of the following system:

$$\dot{\mathbf{z}} = \mathbf{A}\mathbf{z} + \mathbf{B}u$$

$$\mathbf{x} = \mathbf{A}\mathbf{z} + \mathbf{B}u$$

where

$$\mathbf{B} = \mathbf{x}(0), \qquad u = 1(t)$$

Hence a possible MATLAB program for obtaining the response may be given as shown in MATLAB Program 4–10. The resulting response curves are shown in Figure 4–30.

MATLAB Program 4–10

```
A = [0   1;-10   -5];
B = [2;1];
[x,z,t] = step(A,B,A,B);
x1 = [1   0]*x';
x2 = [0   1]*x';
plot(t,x1,'o',t,x2,'-')
grid
title('Response to Initial Condition')
xlabel('t Sec')
ylabel('x1   x2')
```

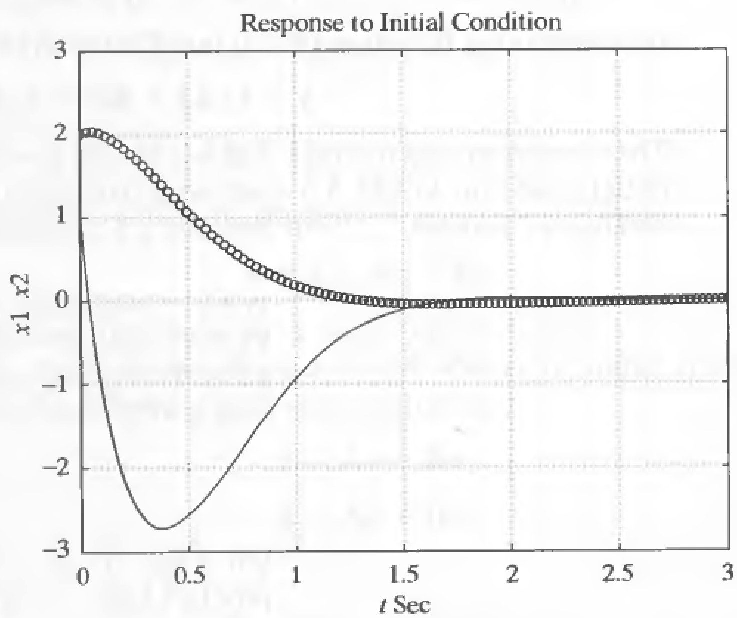


Figure 4–30 Response of system in Example 4–8 to initial condition.

4–5 AN EXAMPLE PROBLEM SOLVED WITH MATLAB

The purpose of this section is to present a MATLAB solution of the response of a mechanical vibratory system. The mathematical model of the system is first developed, then the system is simulated using MATLAB for a continuous-time and a discrete-time approach, and response curves are generated for each approach.

Mechanical vibratory system. Consider the mechanical vibratory system shown in Figure 4–31(a). A wheel has a spring–mass–damper system hanging from it. The wheel is in a track that contains a flat (horizontal) portion, a slanted (downward at 45°)

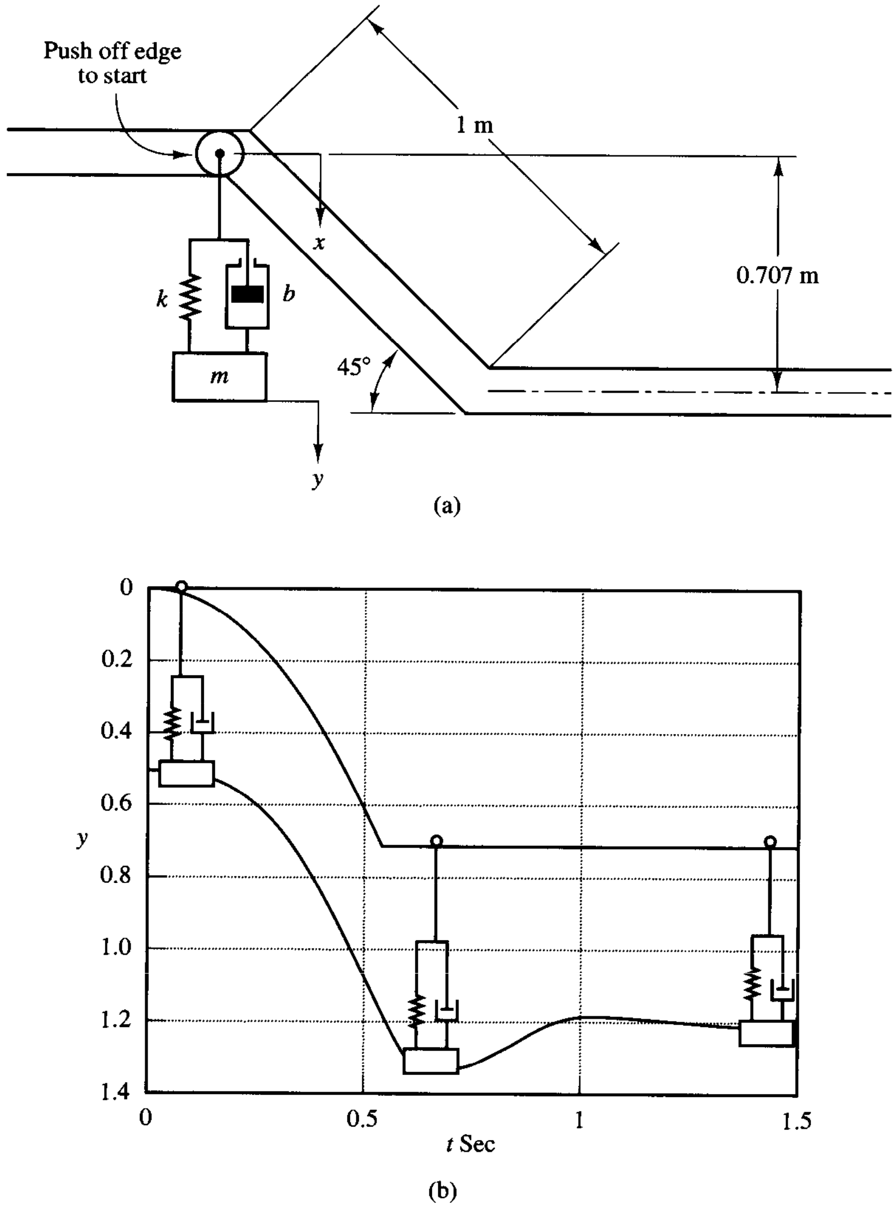


Figure 4–31
(a) Wheel with hanging mass-damper system; (b) dynamic response of the system.

portion, and another flat (horizontal) portion. We start the motion of the system by nudging the wheel over the edge of the ramp. As the wheel drops down the ramp for a total of 0.707 m (vertically measured), the mass m hanging from the spring and damper drops with it, and the mass gains momentum that dissipates gradually. In this problem the wheel is assumed to slide on the slanted portion of the track without friction. On the second flat portion of the track, the wheel slides and rolls. The wheel continues to move on the flat portion of the track until it is stopped by an external means.

Assume the following numerical values for m, b, and k:

$$m = 4 \text{ kg}, \qquad b = 40 \text{ N-sec/m}, \qquad k = 400 \text{ N/m}$$

Assume also that the mass m_p of the wheel is negligible compared with the mass m. Obtain $x(t)$, the vertical motion of the wheel. Then obtain $Y(s)$, the Laplace transform of $y(t)$, which represents the up and down motion of mass m. The coordinate y is attached to the spring–mass–damper system as shown in Figure 4–31 and is measured from the equilibrium position of the system. The initial conditions are that $y(0) = 0$ and $\dot{y}(0) = 0$. Note that in this problem we are interested only in the vertical motions of the spring–mass–damper system. Note also that the system is frictionless with the exception of the damper, which relies on viscosity for its operation.

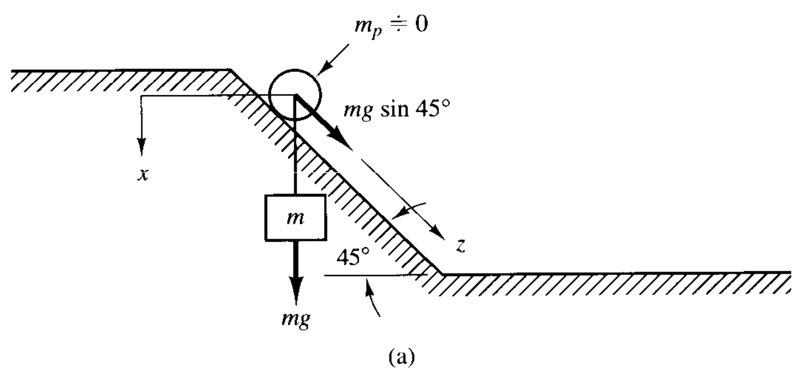
As the spring–mass–damper component travels down the ramp, it will undergo an acceleration produced as a result of the gravity force. When the spring–mass–damper reaches the level region at the bottom of the ramp, a shock will immediately be imposed on the spring–mass–damper component. It will, however, eventually come to a state of equilibrium following the impact due to the settling effects of the damper and spring. The dynamic response of this system is shown in Figure 4–31(b).

Determination of x(t). The system starts with zero initial velocity and follows the track. The input to the system is the vertical position x along the track, and the output is the vertical position y of the mass. Since we assume no sliding friction, referring to Figure 4–32(a) we have in the z direction the following equation:

$$m\ddot{z} = mg \sin 45^\circ$$

or

$$\ddot{z} = 9.81 \times 0.707 = 6.9357$$



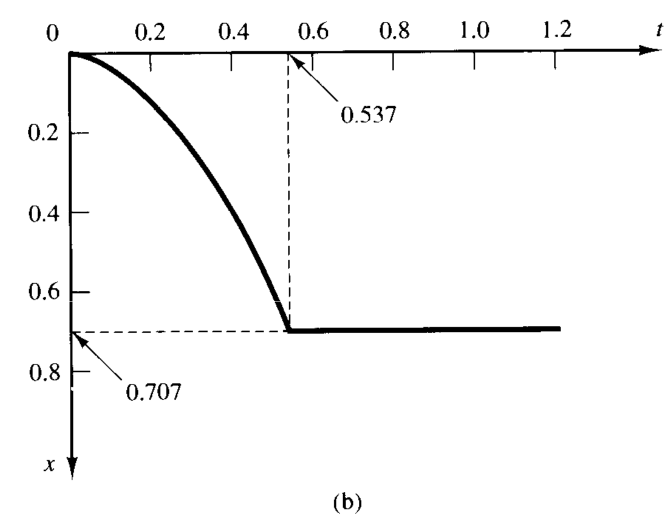


Figure 4–32
(a) Wheel with mass m slides on inclined plane; (b) curve $x(t)$ versus t.

Let us define the time it takes for the wheel to move from $z = 0$ to $z = 1$ m as t_1. Then

$$z = 6.9357 \frac{t_1^2}{2} = 1$$

which yields

$$t_1 = 0.537 \text{ sec}$$

Thus $x(t)$ can be given as follows:

$$\begin{aligned} x(t) = 0.707z = 0.707 \times 3.4678t^2 &= 2.452t^2, \qquad && \text{for } 0 \le t \le 0.537 \\ &= 0.707, && \text{for } 0.537 < t \end{aligned}$$

It follows that from $t = 0.537$ sec to $t = \infty$ we have an input defined by a constant of 0.707. The position x at the end of the ramp is 0.707 and it takes approximately 0.537 sec to get there. A curve $x(t)$ versus t is shown in Figure 4–32(b). Note that the positive direction of $x(t)$ is vertically downward.

To get a better picture of the events taking place in the system, we need to look at the input, shown in Figure 4–32(b). The effects of gravity do not allow us to model the behavior of the system with an ordinary ramp, but rather a parabolic function, which is followed by a constant input.

Determination of Transfer Function $Y(s)/X(s)$. Next, we shall first obtain the equation of motion for the system and then the transfer function $Y(s)/X(s)$. Since y is measured from the equilibrium position, the system equation becomes

$$m\ddot{y} + b(\dot{y} - \dot{x}) + k(y - x) = 0$$

or

$$m\ddot{y} + b\dot{y} + ky = b\dot{x} + kx$$

where x is the input to the system and y is the output. By substituting the given numerical values for m, b, and k, we obtain

$$4\ddot{y} + 40\dot{y} + 400y = 40\dot{x} + 400x$$

or

$$\ddot{y} + 10\dot{y} + 100y = 10\dot{x} + 100x \tag{4–61}$$

The transfer function for the system can now be given by

$$\frac{Y(s)}{X(s)} = \frac{10s + 100}{s^2 + 10s + 100} \tag{4–62}$$

where the input $x(t)$ is given by

$$\begin{aligned} x(t) &= 2.452t^2, \qquad && 0 \le t \le 0.537 \\ &= 0.707, && 0.537 < t \end{aligned} \tag{4–63}$$

Our problem here is to use MATLAB to find the inverse Laplace transform of $Y(s)$ given by Equation (4–62). In what follows we consider two approaches. One is to work in the continuous-time domain using the step command. The other is to work in the discrete-time domain using the filter command. We shall first present the continuous-time approach and then the discrete-time approach.

Computer simulation (continuous-time approach). In the continuous-time approach we separate the time region into two parts; $0 \le t \le 0.537$ and $0.537 < t$.

For $0 \le t \le 0.537$:

$$x_1(t) = 2.452t^2$$

Hence

$$X_1(s) = \frac{2.452 \times 2}{s^3} = \frac{4.904}{s^3}$$

The output $Y(s)$ can then be given by

$$\begin{aligned} Y(s) &= \frac{10s + 100}{s^2 + 10s + 100}\frac{4.904}{s^3} \\ &= \frac{49.04s + 490.4}{s^4 + 10s^3 + 100s^2}\frac{1}{s} \end{aligned} \tag{4–64}$$

For $0.537 < t$:

$$x_2(t) = 0.707$$

Since

$$\frac{Y_2(s)}{X_2(s)} = \frac{10s + 100}{s^2 + 10s + 100}$$

the corresponding differential equation becomes

$$\ddot{y}_2 + 10\dot{y}_2 + 100y_2 = 10\dot{x}_2 + 100x_2$$

The Laplace transform of this last equation becomes

$$\begin{aligned} [s^2Y_2(s) - sy_2(0) - \dot{y}_2(0)] + 10[sY_2(s) - y_2(0)] + 100Y_2(s) \\ = 10[sX_2(s) - x_2(0)] + 100X_2(s) \end{aligned}$$

or

$$\begin{aligned} (s^2 + 10s + 100)Y_2(s) = (10s + 100)X_2(s) + sy_2(0) \\ + \dot{y}_2(0) + 10y_2(0) - 10x_2(0) \end{aligned}$$

Hence

$$Y_2(s) = \frac{10s + 100}{s^2 + 10s + 100} X_2(s) + \frac{sy_2(0) + \dot{y}_2(0) + 10y_2(0) - 10x_2(0)}{s^2 + 10s + 100}$$

The initial conditions are found from $y_2(0) = y_1(0.537)$ and $\dot{y}_2(0) = \dot{y}_1(0.537)$. Therefore,

$$Y_2(s) = \frac{10s + 100}{s^2 + 10s + 100}\frac{0.707}{s} + \frac{s^2[y_1(0.537)] + [\dot{y}_1(0.537) + 10y_1(0.537) - 10(0.707)]s}{s^2 + 10s + 100}\frac{1}{s}$$

or

$$Y_2(s) = \frac{10s + 100}{s^2 + 10s + 100}\frac{0.707}{s} + \frac{s^2[y1(537)] + [y1dot(537) + 10y1(537) - 7.07]s}{s^2 + 10s + 100}\frac{1}{s} \tag{4-65}$$

where

$$y1(537) = y_1(0.537), \qquad y1dot(537) = \dot{y}_1(0.537)$$

A MATLAB program to obtain the response $y(t)$ based on the continuous-time approach is given in MATLAB Program 4–11. The resulting response curve $y(t)$ versus t, as well as the input $x(t)$ versus t, is shown in Figure 4–33.

MATLAB Program 4–11

```
% ---------- Continuous-time approach ----------

% ***** Obtain y1 and y1dot *****

num1 = [0   0   0   49.04   490.4];
den1 = [1   10   100   0   0];
t1 = 0:0.001:0.537;
y1 = step(num1,den1,t1);
num2 = [0   0   49.04   490.4];
den2 = [1   10   100   0];
y1dot = step(num2,den2,t1);

% ***** Determine the initial values of y1(537) and y1dot(537)
% for the second part. The initial values for the second
% part are output y2_ini = y1(537) and y2dot_ini = y1dot(537) *****
```

```
y2_ini = y1(537);
y2dot_ini = y1dot(537);
t2 = 0.538:0.001:1.5;
num3 = [0   7.07   70.7];
den3 = [1   10   100];
num4 = [y1(537)   y1dot(537)+10*y1(537)-7.07   0];
y2o = step(num3,den3,t2);
y2i = step(num4,den3,t2);
y2 = y2o + y2i;
y = [y1'   y2'];
t = [t1   t2];
plot(t,-y,'.')

hold

Current plot held

x1 = 2.452*t1.^2;
x2 = 0.707*ones(size(t2));
x = [x1   x2];
plot(t,-x,'-')
grid
title('Response of System (Continuous-Time Approach)')
xlabel('t Sec')
ylabel('Input x and Output y')
text(0.2,-0.54,'Input x')
text(0.47,-0.25,'Output y')
```

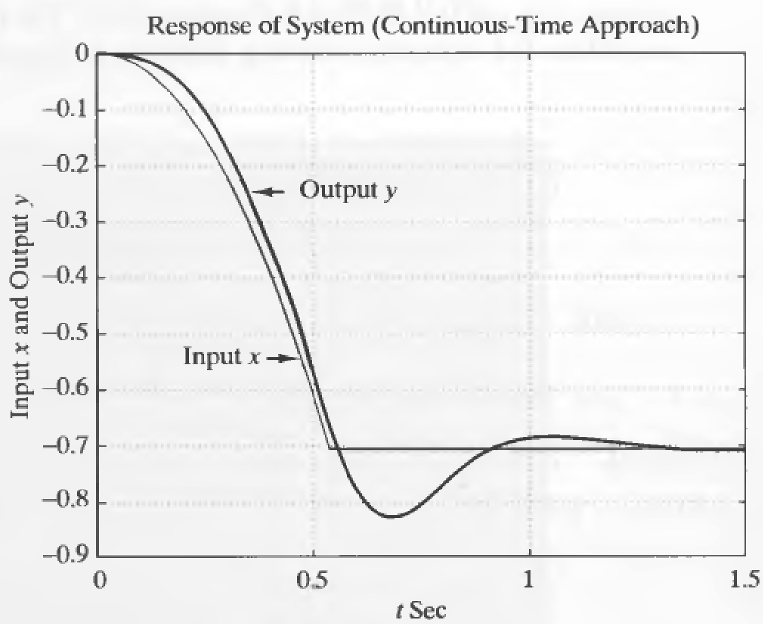


Figure 4–33
Input $x(t)$ and output $y(t)$ obtained by the continuous-time approach.

It is noted that when plotting multiple curves on one diagram we may use the "hold" command. If we enter the command "hold" in the computer, the screen will show

```
hold

Current plot held
```

To release the held plot, enter the command "hold" again. Then the current plot will be released as shown next.

```
hold

Current plot held

hold

Current plot released
```

Computer simulation (discrete-time approach). The continuous-time transfer function may be converted to a pulse transfer function (discrete-time transfer function) using general formulas. The simpler method is to convert the continuous-time transfer function to a pulse transfer function using MATLAB commands. The first step is to convert the continuous-time transfer function to a continuous-time set of state-space equations using the MATLAB command [A,B,C,D] = tf2ss(num,den). The state-space equations can then be converted from continuous-time to discrete-time using the command [G,H] = c2d(A,B,Ts), where Ts is the desired time step (sampling period). The discrete-time state-space equations are converted to a pulse transfer function with the command [numz,denz] = ss2tf(G,H,C,D).

In the present case we choose $T = 0.001$ sec. The input function $x(t)$ must first be discretized. The continuous-time input function was determined to be

$$x(t) = 2.452t^2, \qquad \text{for } 0 \leqslant t \leqslant 0.537$$

$$x(t) = 0.707, \qquad \text{for } 0.537 < t$$

Note that we define x as an array of points in MATLAB. This array initially follows $x(t) = 2.452t^2$ and, after $t = 0.537$ sec, follows $x(t) = 0.707$. We assume that the time region is $0 \leqslant t \leqslant 1.5$.

The acceleration input in the first part can be written as

```
k1 = 0:537;
x1 = [2.452*(0.001*k1).^2]
```

where k1 represents a time count and x1 is the first part of the complete input function. (There are 538 calculation points from the initial position until the input reaches

0.707 m.) For the second part of the input, we need a step function with magnitude 0.707. After time 0.537 sec,

```
k2 = 538:1500;
x2 = [0.707*ones(size(k2))]
```

(There are 963 points from 0.538 sec through 1.5 sec, inclusive.) The next step is to transform both inputs to one complete input:

```
x = [x1    x2];
```

(The two input equations are transformed into a single vector in order to appear as a single entry in the filter command argument.)

Now we can use the filter command, assigning a variable y,

```
y = filter(numz,denz,x);
```

and plot the response y(t), as well as the original input itself, x(t), taking care with the time intervals using t:

```
t = 0:1500;
plot(t/1000,-y,'.',t/1000,-x,'-')
```

(We divide t by 1000 because the time step is 0.001 sec.) Note also that the plotted input and output functions are negated. (Otherwise, we would have a positive acceleration input and response, which would be incorrect.)

A possible MATLAB program using the discrete-time approach is shown in MATLAB Program 4–12. The resulting response curves $x(t)$ versus t and $y(t)$ versus t are shown in Figure 4–34.

MATLAB Program 4–12

```
% ---------- Discrete-time approach ----------

% ***** Convert continuous-time transfer function to discrete-time
% transfer function (pulse transfer function) by choosing the time
% step (sampling period) to be 0.001 sec *****

num = [0   10   100];
den = [1   10   100];
[A,B,C,D] = tf2ss(num,den);
[G,H] = c2d(A,B,0.001);
[numz,denz] = ss2tf(G,H,C,D);

% ***** Enter the inputs x1 and x2 *****
```

```
k1 = 0:537;
x1 = [2.452*(0.001*k1).^2];
k2 = 538:1500;
x2 = [0.707*ones(size(k2))];
x = [x1   x2];

% ***** Using the filter command, obtain the response y *****

y = filter(numz,denz,x);
t = 0:1500;
plot(t/1000,-y,'.',t/1000,-x,'-')
grid
title('Response of System (Discrete-Time Approach)')
xlabel('t Sec')
ylabel('Input x  and  Output y')
text(0.2,-0.54,'Input x')
text(0.47,-0.25, 'Output y')
```

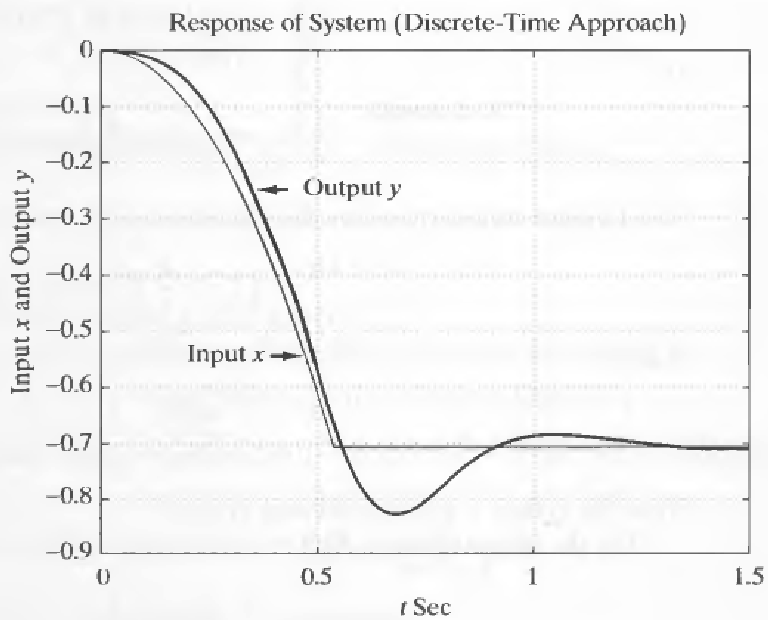


Figure 4–34 Input $x(t)$ and output $y(t)$ obtained by the discrete-time approach.

EXAMPLE PROBLEMS AND SOLUTIONS

A-4-1. In the system of Figure 4–35, $x(t)$ is the input displacement and $\theta(t)$ is the output angular displacement. Assume that the masses involved are negligibly small and that all motions are restricted to be small; therefore, the system can be considered linear. The initial conditions for x and θ are zeros, or $x(0-) = 0$ and $\theta(0-) = 0$. Show that this system is a differentiating element. Then obtain the response $\theta(t)$ when $x(t)$ is a unit-step input.

Solution. The equation for the system is

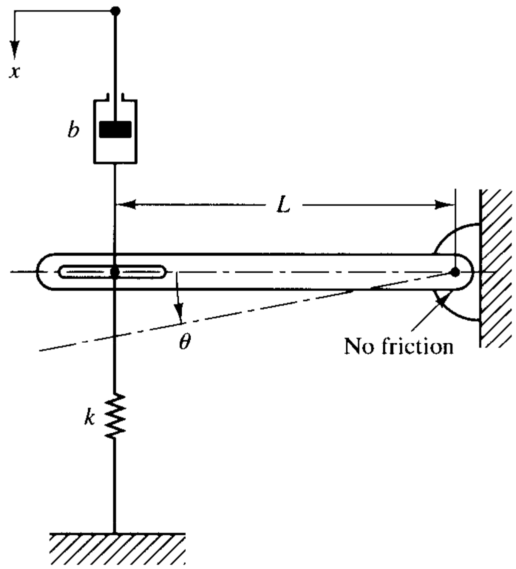


Figure 4–35
Mechanical system.

$$b(\dot{x} - L\dot{\theta}) = kL\theta$$

or

$$L\dot{\theta} + \frac{k}{b}L\theta = \dot{x}$$

The Laplace transform of this last equation, using zero initial conditions, gives

$$\left(Ls + \frac{k}{b}L\right)\Theta(s) = sX(s)$$

And so

$$\frac{\Theta(s)}{X(s)} = \frac{1}{L}\frac{s}{s + (k/b)}$$

Thus the system is a differentiating system.

For the unit-step input $X(s) = 1/s$, the output $\Theta(s)$ becomes

$$\Theta(s) = \frac{1}{L}\frac{1}{s + (k/b)}$$

The inverse Laplace transform of $\Theta(s)$ gives

$$\theta(t) = \frac{1}{L}e^{-(k/b)t}$$

Note that if the value of k/b is large the response $\theta(t)$ approaches a pulse signal as shown in Figure 4–36.

A-4-2. Consider the mechanical system shown in Figure 4–37. Suppose that the system is at rest initially $[x(0) = 0, \dot{x}(0) = 0]$, and at $t = 0$ it is set into motion by a unit-impulse force. Obtain a mathematical model for the system. Then find the motion of the system.

Solution. The system is excited by a unit-impulse input. Hence

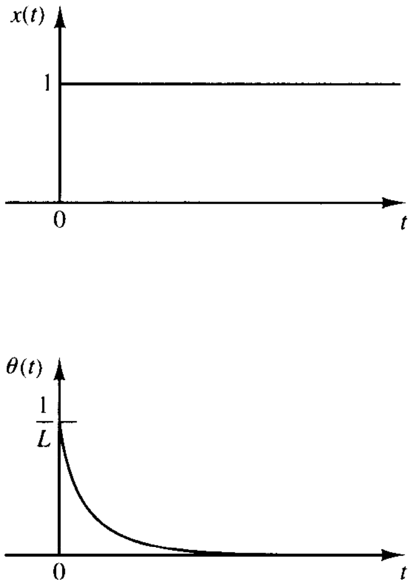


Figure 4–36
Unit-step input and the response of the mechanical system shown in Figure 4–35.

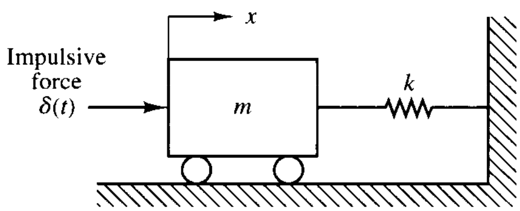


Figure 4–37
Mechanical system.

$$m\ddot{x} + kx = \delta(t)$$

This is a mathematical model for the system.

Taking the Laplace transform of both sides of this last equation gives

$$m[s^2X(s) - sx(0) - \dot{x}(0)] + kX(s) = 1$$

By substituting the initial conditions $x(0) = 0$ and $\dot{x}(0) = 0$ into this last equation and solving for $X(s)$, we obtain

$$X(s) = \frac{1}{ms^2 + k}$$

The inverse Laplace transform of $X(s)$ becomes

$$X(t) = \frac{1}{\sqrt{mk}} \sin \sqrt{\frac{k}{m}}\, t$$

The oscillation is a simple harmonic motion. The amplitude of the oscillation is $1/\sqrt{mk}$.

A-4-3. Obtain the closed-loop transfer function for the positional servo system shown in Figure 4–38. Assume that the input and output of the system are the input shaft position and the output shaft position, respectively. Assume the following numerical values for system constants:

r = angular displacement of the reference input shaft, radians
c = angular displacement of the output shaft, radians
θ = angular displacement of the motor shaft, radians
K_0 = gain of the potentiometric error detector = $24/\pi$ V/rad

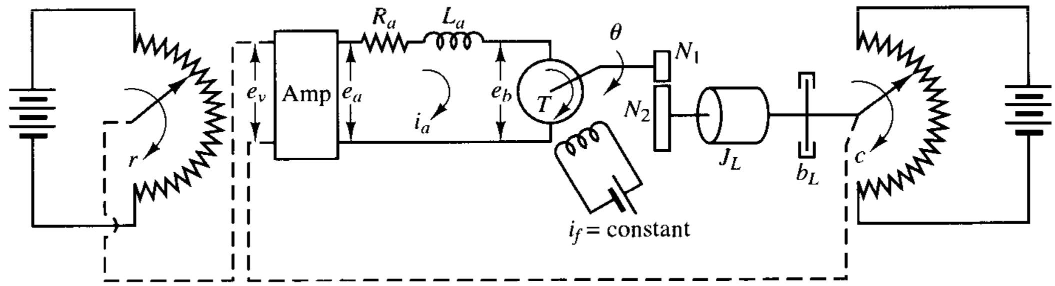


Figure 4–38
Positional servo system.

K_1 = amplifier gain = 10 V/V
e_a = armature voltage, V
e_b = back emf, V
R_a = armature-winding resistance = 0.2 Ω
L_a = armature-winding inductance = negligible
i_a = armature-winding current, A
K_3 = back emf constant = 5.5×10^{-2} V-sec/rad
K_2 = motor torque constant = 6×10^{-5} N-m/A
J_m = moment of inertia of the motor referred to the motor shaft = 1×10^{-5} kg-m^2
b_m = viscous-friction coefficient of the motor referred to the motor shaft = negligible
J_L = moment of inertia of the load referred to the output shaft = 4.4×10^{-3} kg-m^2
b_L = viscous-friction coefficient of the load referred to the output shaft = 4×10^{-2}N-m/rad/sec
n = gear ratio $N_1/N_2 = \frac{1}{10}$

Solutiou. The equivalent moment of inertia J_0 and equivalent viscous friction coefficient b_0 referred to the motor shaft are, respectively,

$$J_0 = J_m + n^2 J_L$$
$$= 1 \times 10^{-5} + 4.4 \times 10^{-5} = 5.4 \times 10^{-5}$$
$$b_0 = b_m + n^2 b_L$$
$$= 4 \times 10^{-4}$$

Referring to Equation (4–16), we obtain

$$\frac{C(s)}{E(s)} = \frac{K_m}{s(T_m s + 1)}$$

where

$$K_m = \frac{K_0 K_1 K_2 n}{R_a b_0 + K_2 K_3} = \frac{7.64 \times 10 \times 6 \times 10^{-5} \times 0.1}{(0.2)(4 \times 10^{-4}) + (6 \times 10^{-5})(5.5 \times 10^{-2})} = 5.5$$

$$T_m = \frac{R_a J_0}{R_a b_0 + K_2 K_3} = \frac{(0.2)(5.4 \times 10^{-5})}{(0.2)(4 \times 10^{-4}) + (6 \times 10^{-5})(5.5 \times 10^{-2})} = 0.13$$

Thus,

$$\frac{C(s)}{E(s)} = \frac{5.5}{s(0.13s + 1)} \tag{4–66}$$

Using Equation (4–66), we can draw the block diagram of the system as shown in Figure 4–39. The closed-loop transfer function of the system is

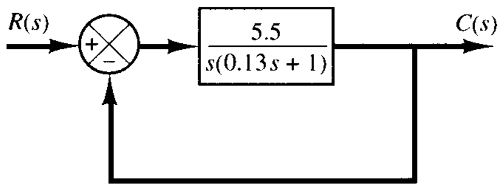


Figure 4–39
Block diagram of the system shown in Figure 4–38.

$$\frac{C(s)}{R(s)} = \frac{5.5}{0.13s^2 + s + 5.5} = \frac{42.3}{s^2 + 7.69s + 42.3}$$

A-4-4. Gear trains are often used in servo systems to reduce speed, to magnify torque, or to obtain the most efficient power transfer by matching the driving member to the given load.

Consider the gear train system shown in Figure 4–40. In this system, a load is driven by a motor through the gear train. Assuming that the stiffness of the shafts of the gear train is infinite (there is neither backlash nor elastic deformation) and that the number of teeth on each gear is proportional to the radius of the gear, obtain the equivalent moment of inertia and equivalent viscous-friction coefficient referred to the motor shaft and referred to the load shaft.

In Figure 4–40 the numbers of teeth on gears 1, 2, 3, and 4 are N_1, N_2, N_3, and N_4, respectively. The angular displacements of shafts, 1, 2, and 3 are θ_1, θ_2, and θ_3, respectively. Thus, $\theta_2/\theta_1 = N_1/N_2$ and $\theta_3/\theta_2 = N_3/N_4$. The moment of inertia and viscous-friction coefficient of each gear train component are denoted by J_1, b_1; J_2, b_2; and J_3, b_3; respectively. (J_3 and b_3 include the moment of inertia and friction of the load.)

Solution. For this gear train system, we can obtain the following three equations: For shaft 1,

$$J_1\ddot{\theta}_1 + b_1\dot{\theta}_1 + T_1 = T_m \tag{4–67}$$

where T_m is the torque developed by the motor and T_1 is the load torque on gear 1 due to the rest of the gear train. For shaft 2,

$$J_2\ddot{\theta}_2 + b_2\dot{\theta}_2 + T_3 = T_2 \tag{4–68}$$

where T_2 is the torque transmitted to gear 2 and T_3 is the load torque on gear 3 due to the rest of the gear train. Since the work done by gear 1 is equal to that of gear 2,

$$T_1\theta_1 = T_2\theta_2 \qquad \text{or} \qquad T_2 = T_1\frac{N_2}{N_1}$$

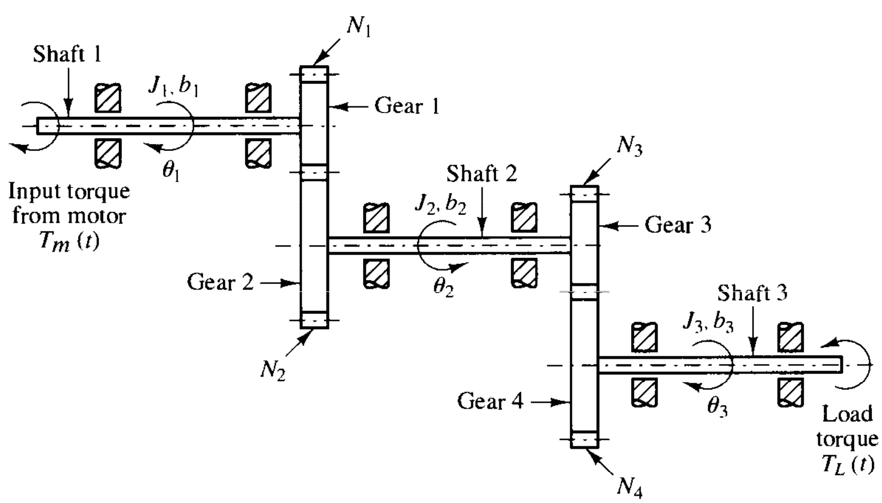


Figure 4–40
Gear train system.

If $N_1/N_2 < 1$, the gear ratio reduces the speed as well as magnifies the torque. For the third shaft,

$$J_3\ddot{\theta}_3 + b_3\dot{\theta}_3 + T_L = T_4 \tag{4–69}$$

where T_L is the load torque and T_4 is the torque transmitted to gear 4. T_3 and T_4 are related by

$$T_4 = T_3 \frac{N_4}{N_3}$$

and θ_3 and θ_1 are related by

$$\theta_3 = \theta_2 \frac{N_3}{N_4} = \theta_1 \frac{N_1}{N_2} \frac{N_3}{N_4}$$

Elimination of T_1, T_2, T_3, and T_4 from Equations (4–67), (4–68), and (4–69) yields

$$J_1\ddot{\theta}_1 + b_1\dot{\theta}_1 + \frac{N_1}{N_2}(J_2\ddot{\theta}_2 + b_2\dot{\theta}_2) + \frac{N_1N_3}{N_2N_4}(J_3\ddot{\theta}_3 + b_3\dot{\theta}_3 + T_L) = T_m$$

Eliminating θ_2 and θ_3 from this last equation and writing the resulting equation in terms of θ_1 and its time derivatives, we obtain

$$\left[J_1 + \left(\frac{N_1}{N_2}\right)^2 J_2 + \left(\frac{N_1}{N_2}\right)^2 \left(\frac{N_3}{N_4}\right)^2 J_3\right]\ddot{\theta}_1 + \left[b_1 + \left(\frac{N_1}{N_2}\right)^2 b_2 + \left(\frac{N_1}{N_2}\right)^2 \left(\frac{N_3}{N_4}\right)^2 b_3\right]\dot{\theta}_1 + \left(\frac{N_1}{N_2}\right)\left(\frac{N_3}{N_4}\right)T_L = T_m \tag{4–70}$$

Thus, the equivalent moment of inertia and viscous-friction coefficient of the gear train referred to shaft 1 are given, respectively, by

$$J_{1\text{eq}} = J_1 + \left(\frac{N_1}{N_2}\right)^2 J_2 + \left(\frac{N_1}{N_2}\right)^2 \left(\frac{N_3}{N_4}\right)^2 J_3$$

$$b_{1\text{eq}} = b_1 + \left(\frac{N_1}{N_2}\right)^2 b_2 + \left(\frac{N_1}{N_2}\right)^2 \left(\frac{N_3}{N_4}\right)^2 b_3$$

Similarly, the equivalent moment of inertia and viscous-friction coefficient of the gear train referred to the load shaft (shaft 3) are given, respectively, by

$$J_{3\text{eq}} = J_3 + \left(\frac{N_4}{N_3}\right)^2 J_2 + \left(\frac{N_2}{N_1}\right)^2 \left(\frac{N_4}{N_3}\right)^2 J_1$$

$$b_{3\text{eq}} = b_3 + \left(\frac{N_4}{N_3}\right)^2 b_2 + \left(\frac{N_2}{N_1}\right)^2 \left(\frac{N_4}{N_3}\right)^2 b_1$$

The relationship between $J_{1\text{eq}}$ and $J_{3\text{eq}}$ is thus

$$J_{1\text{eq}} = \left(\frac{N_1}{N_2}\right)^2 \left(\frac{N_3}{N_4}\right)^2 J_{3\text{eq}}$$

and that between $b_{1\text{eq}}$ and $b_{3\text{eq}}$ is

$$b_{1\text{eq}} = \left(\frac{N_1}{N_2}\right)^2 \left(\frac{N_3}{N_4}\right)^2 b_{3\text{eq}}$$

The effect of J_2 and J_3 on an equivalent moment of inertia is determined by the gear ratios N_1/N_2 and N_3/N_4. For speed-reducing gear trains, the ratios N_1/N_2 and N_3/N_4 are usually less than unity.